AMG CONTROL PHILOSOPHY

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2. Introduction.

This document describes the operating cycles and sequences of the Elevator Conveyor System, supplied by Conveyor Logistic for the AMG plant.

It also outlines the guiding principles and control strategies for the operation, monitoring, and safety of the Freezer Pallet Elevators Control System.

3. Glossary

- PLC: a programmable logic controller is an industrial computer control system that continuously monitors the state of input devices and makes decisions based upon a custom program to control the state of output devices.

- VSD: a variable speed drives controls the speed and torque of an AC motor by converting fixed frequency and voltage input to a variable frequency and voltage output. System performance can be greatly improved by controlling speed to precisely match the load.

- HMI: A Human Machine Interface is a user interface or dashboard that connects a person to a machine, system, or device. While the term can technically be applied to any screen that allows a user to interact with a device.

- Safety Relays: Safety Relays are devices that implement safety functions. In the event of a hazard, a safety relay will work to reduce to risk to an acceptable level. When an error occurs, the safety relay will initiate a safe and reliable response.

- Photo electric sensor: a photoelectric sensor emits a light beam (visible or infrared) from its light-emitting element. A reflective-type photoelectric sensor is used to detect the light beam reflected from the target.

- 3PH: Three-phase electric power is a common method of alternating current electric power generation, transmission, and distribution. It is a type of polyphase system and is the most common method used to power large motors and other heavy loads.

- DOL. A direct on-line is the simplest type of motor starter. When energised, a direct on-line starter immediately connects the motor terminals directly to the power supply. A DOL motor starter often contains protection devices, and in some cases, condition monitoring. Those requiring remote or automatic control uses an electromechanical contactor to switch the motor circuit.

4. Brief functioning description

The primary function of the system is to transport pallets between the ground level and various storage levels as specified by the Warehouse Management System (WMS). This includes moving pallets from ground level to designated levels and returning stored pallets back to the ground.

The handling system consists of two Elevator Conveyor subsystems, each with the following main components:

Elevator A Pallet Conveyor Subsystem:

- Ground-Level Pallet Chain Conveyors (4 units). These conveyors handle the transportation, loading, and unloading of pallets from Elevator A at ground level:
  - F_PCC_01A.
  - F_PCC_02A.
  - F_PCC_03A.
  - F_PCC_05A.

- Ground-Level Transfer Conveyor (1 unit). Handles pallet movement and direction changes at the infeed/outfeed point of Elevator A:
  - F_PCC_04A.

- Ground-Level Dimensioner (1 unit). Measures the dimensions of pallet loads to prevent collisions and obstructions, ensuring operational safety for both the system and personnel.

- Ground-Level Roller Door (1 unit). These doors help maintain the internal temperature of the storage area by opening only when pallet traffic is detected and closing automatically when there is no activity.

- Lift Carrier A (1 unit). A mechanical subsystem that positions the conveyor within Elevator A's carrier platform.

- Lift Carrier Pallet Chain Conveyor (1 unit). Moves, loads, and unloads pallets within Elevator A across all levels:
  - F_PCC_LIFT_A.

- Elevator A (1 unit). Responsible for the vertical movement of pallets between floors.

- Level Pallet Chain Conveyors with Shuttle Access (9 units). These subsystems conveyors receive pallets at each level from the elevator and transfer them to the sorting robot for processing. Also, depending on the configuring direction at the HMI these conveyors can receive pallets after sorting. Once loaded by the sorting robot, pallets are transferred to the elevator.
  - F_PCC_WSA_01A.

  - F_PCC_WSA_02A.
  - F_PCC_WSA_03A.
  - F_PCC_WSA_04A.
  - F_PCC_WSA_05A.
  - F_PCC_WSA_06A.
  - F_PCC_WSA_07A.
  - F_PCC_WSA_08A.
  - F_PCC_WSA_09A.

Elevator B Pallet Conveyor Subsystem:

- Ground-Level Pallet Chain Conveyors (5 units). These conveyors manage the transportation, loading, and unloading of pallets from Elevator B at ground level:
  - F_PCC_02B.
  - F_PCC_03B.
  - F_PCC_04B.
  - F_PCC_05B.
  - F_PCC_06B.

- Ground-Level Transfer Conveyor (1 unit). Handles pallet movement and direction changes at the infeed/outfeed point of Elevator B:
  - F_PCC_01B.

- Ground-Level Dimensioner (1 unit). Measures the dimensions of pallet loads to prevent collisions and obstructions, ensuring operational safety for both the system and personnel.

- Ground-Level Roller Door (1 unit). These doors help maintain the internal temperature of the storage area by opening only when pallet traffic is detected and closing automatically when there is no activity.

- Lift Carrier A (1 unit). A mechanical subsystem that positions the conveyor within Elevator A's carrier platform.

- Lift Carrier Pallet Chain Conveyor (1 unit). Moves, loads, and unloads pallets within Elevator A across all levels:
  - F_PCC_LIFT_A.

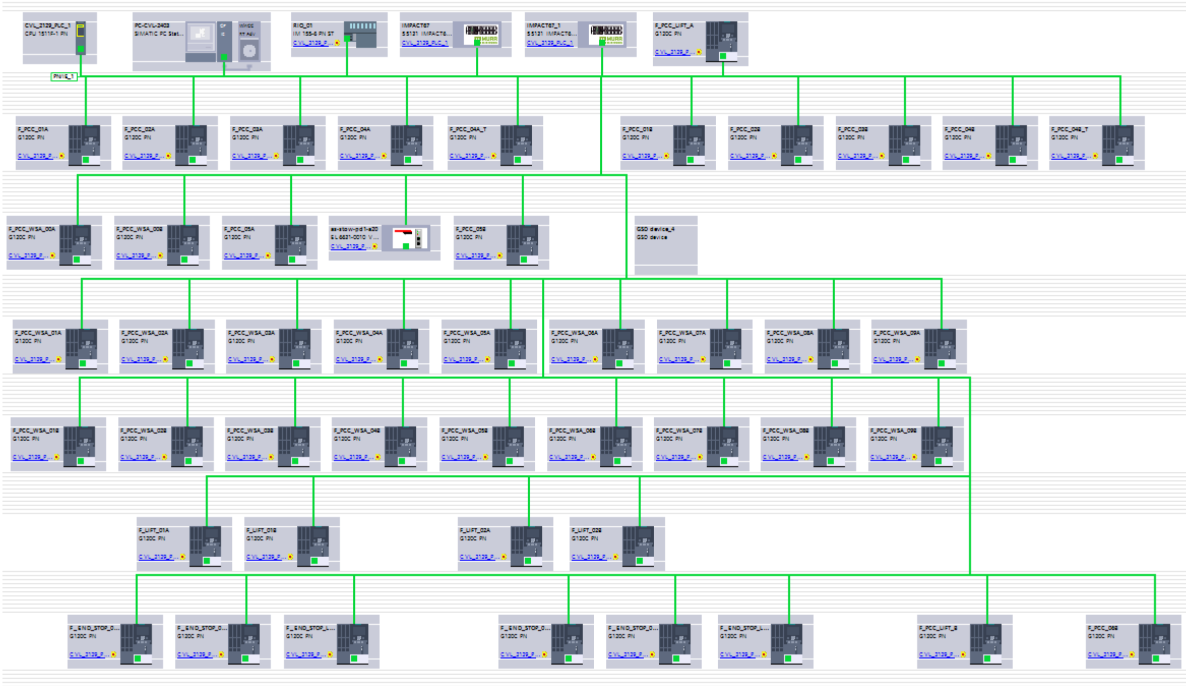
- Elevator A (1 unit). Responsible for the vertical movement of pallets between floors.

- Level Pallet Chain Conveyors with Shuttle Access (9 units). These subsystems conveyors receive pallets at each level from the elevator and transfer them to the sorting robot for processing. Also, depending on the configuring direction at the HMI these conveyors can receive pallets after sorting. Once loaded by the sorting robot, pallets are transferred to the elevator.
  - F_PCC_WSA_01B.
  - F_PCC_WSA_02B.

- F_PCC_WSA_03B.
- F_PCC_WSA_04B.
- F_PCC_WSA_05B.
- F_PCC_WSA_06B.
- F_PCC_WSA_07B.
- F_PCC_WSA_08B.
- F_PCC_WSA_09B.

5. Control System Architecture

The control system is based on a Siemens S7-1500 Safety PLC, which governs the core logic of the pallet handling operation. Utilizing PROFINET communication, it controls and monitors various peripherals and variable speed drives (VSDs), including actuators, motors, and sensors integrated into the distribution panels.



5.1. Field devices communicating via Profinet

The following field devices are connected and managed via PROFINET:

- 45 Siemens G120C PN Variable Speed Drives (VSDs): These drives regulate and monitor AC motor speed, torque, and other parameters by converting a fixed input frequency and voltage to a variable output. The drives are fully automated and controlled via the PLC.

- 1 Siemens IM 155-6 PN ST Remote I/O (Coupler): Serves as the main interface for the system's inputs and outputs. It enables the PLC to control and monitor field devices such as sensors, DOL motors, and actuators.

- 2 Murr Elektronik Remote I/Os (Couplers): These are used specifically for reading the elevator level sensors. They transmit sensor status and measurements to the PLC.

- 1 Beckhoff PROFINET-to-EtherCAT Gateway: Facilitates communication between the Siemens PLC and the Stow system, managing the exchange of pallet-related data at each storage level.

5.2. Human Machine Interface.

Operator interaction with the system occurs through the Human-Machine Interface (HMI), which runs on an Industrial PC (IPC) using Siemens WinCC software. The HMI serves as a robust, user-friendly interface for monitoring and controlling the entire system. It exchanges real-time data with the PLC over the PROFINET network.

The HMI enables:

- Manual and semi-automatic control of conveyor systems
- Monitoring system status and alarms
- Parameter adjustments
- Diagnostics and maintenance access

5.3. Control level.

The control architecture follows industrial automation standards, with the **PLC** maintaining central control and logic processing. The system supports two operational modes:

- **Local Control:**
  Initiated directly via the HMI. This includes manual and semi-automatic sequences, often used for maintenance or troubleshooting.

- **Remote (Automatic) Control:**
  Operated through instructions received from the **WMS (Warehouse Management System)**. In this mode, the PLC executes fully automated movements based on WMS commands.

5.4. Communication networks and protocols

The system employs the following communication protocols:

- **PROFINET (Process Field Network):**
  An industrial Ethernet protocol used for real-time communication between the PLC and field devices (e.g., sensors, actuators, VSDs).
- **JSON over TCP/IP Socket:**
  A lightweight custom protocol used for communication between the PLC and the WMS. Data is exchanged in **JSON format** over a TCP/IP socket, enabling efficient and structured message transmission.
- **TCP/IP Socket for Barcode Scanners:**
  Barcode scanners communicate via TCP/IP sockets. Each time a scan is triggered, the scanner sends a text string directly to the PLC for processing.

6. Description of the subsystem equipment.

The subsystem comprises a variety of equipment, each with its own purpose and operation. This includes actuator, motors, and other related components.

6.1. Gound level pallet chain conveyor-Elevator A/B.

These conveyors serve as the entry and exit points of the system, facilitating pallet movement between ground level and the elevator system. Functionally, they handle the following:

- **Infeed Operation:** Pallets enter the system and are transported through the dimensioner, where their dimensions are evaluated. Once verified, the pallets are conveyed to the elevator to be transported to their designated storage level.
- **Outfeed Operation:** If configured for outfeed via the HMI, this conveyor section receives pallets from the elevator and transports them to the system exit.
- **Dynamic Configuration:** The operational direction (infeed or outfeed) is configurable via the HMI, allowing operators to modify the flow based on operational needs.

System Components:

- **Sensors (2 per conveyor):** These detect the presence and position of pallets. Depending on the configured direction of travel, they inform the **PLC** whether the zone is occupied or ready for a new pallet.
- **Three-Phase Motor with VSD (Variable Speed Drive):** Each conveyor is equipped with a motor and VSD unit. The **VSD** receives commands from the **PLC** via **PROFINET** to control speed, direction, and motor operation, ensuring smooth pallet transfer between conveyor zones.
- **Light Curtains:** Light curtains are safety sensors that detect unauthorized or unexpected presence at the system's entrance or exit. Their primary function is to halt conveyor operation if any external presence is detected, ensuring safe system operation.
- **Start/Reset Push Button:** This button is used to reset the conveyors and restart their movement after the light curtain has been triggered.

The following sections detail the specific devices used within each conveyor zone.

6.1.1. F_PCC_01A- Gound level pallet chain conveyor.

F_PCC_01A (Elevator A). Pallet infeed/outfeed chain conveyor.

Device	Tag	Description
PE Sensor 1	1.1A-SEN.001	Front sensor used when the conveyor is loading pallets onto Elevator A. It switches to the rear position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PE Sensor 2	2.1A-SEN.002	Rear sensor used when the conveyor is loading pallets onto Elevator A. It switches to the front position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
Motor-VSD	3.F_PCC_01A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.
Light Curtains	4. LIGTHC_A-SEN.001	Light curtains are security sensors that detect unauthorized or unexpected presence at the entrance or exit of the system.
Start/Reset Push Button	5. START_RST-PB.001	This button is used to reset the conveyors and restart their movement after the light curtain has been triggered.

6.1.2. F_PCC_02A- Gound level pallet chain conveyor.

F_PCC_02A (Elevator A). Pallet infeed/outfeed chain conveyor.

Device	Tag	Description
PE Sensor 1	6.2A-SEN.001	Front sensor used when the conveyor is loading pallets onto Elevator A. It switches to the rear position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PE Sensor 2	7.2A-SEN.002	Rear sensor used when the conveyor is loading pallets onto Elevator A. It switches to the front position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
Motor-VSD	8.F_PCC_02A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.1.3. F_PCC_03A- Gound level pallet chain conveyor.

F_PCC_03A (Elevator A). Pallet infeed/outfeed chain conveyor.

Device	Tag	Description
PE Sensor 1	9.3A-SEN.001	Front sensor used when the conveyor is loading pallets onto Elevator A. It switches to the rear position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PE Sensor 2	10.3A-SEN.002	Rear sensor used when the conveyor is loading pallets onto Elevator A. It switches to the front position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
Motor-VSD	11.F_PCC_03A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.1.4. F_PCC_05A- Gound level pallet chain conveyor.

F_PCC_05A (Elevator A). Pallet infeed/outfeed chain conveyor.

Device	Tag	Description
PE Sensor 1	12.5A-SEN.001	Front sensor used when the conveyor is loading pallets onto Elevator A. It switches to the rear position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PE Sensor 2	13.5A-SEN.002	Rear sensor used when the conveyor is loading pallets onto Elevator A. It switches to the front position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
Motor-VSD	14.F_PCC_05A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.1.5. F_PCC_02B- Gound level pallet chain conveyor.

F_PCC_02B (Elevator B). Pallet outfeed/infeed chain conveyor.

Device	Tag	Description
PE Sensor 1	15.2B-SEN.001	Front sensor used when the conveyor is unloading pallets from Elevator B. It switches to the rear position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PE Sensor 2	16.2B-SEN.002	Rear sensor used when the conveyor is unloading pallets from Elevator B. It switches to the front position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
Motor-VSD	17.F_PCC_02B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.1.6. F_PCC_03B- Gound level pallet chain conveyor.

F_PCC_03B (Elevator B). Pallet outfeed/infeed chain conveyor.

Device	Tag	Description
PE Sensor 1	18.3B-SEN.001	Front sensor used when the conveyor is unloading pallets from Elevator B. It switches to the rear position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PE Sensor 2	19.3B-SEN.002	Rear sensor used when the conveyor is unloading pallets from Elevator B. It switches to the front position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
Motor-VSD	20.F_PCC_03B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.1.7. F_PCC_04B- Gound level pallet chain conveyor.

F_PCC_04B (Elevator B). Pallet outfeed/infeed chain conveyor.

Device	Tag	Description
PE Sensor 1	21.4B-SEN.001	Front sensor used when the conveyor is unloading pallets from Elevator B. It switches to the rear position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PE Sensor 2	22.4B-SEN.002	Rear sensor used when the conveyor is unloading pallets from Elevator B. It switches to the front position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
Motor-VSD	23.F_PCC_04B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.1.8. F_PCC_05B- Gound level pallet chain conveyor.

F_PCC_05B (Elevator B). Pallet outfeed/infeed chain conveyor.

Device	Tag	Description
PE Sensor 1	24.5B-SEN.001	Front sensor used when the conveyor is unloading pallets from Elevator B. It switches to the rear position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PE Sensor 2	25.5B-SEN.002	Rear sensor used when the conveyor is unloading pallets from Elevator B. It switches to the front position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
Motor-VSD	26.F_PCC_05B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.1.9. F_PCC_06B- Gound level pallet chain conveyor.

F_PCC_06B (Elevator B). Pallet outfeed/infeed chain conveyor.

Device	Tag	Description
PE Sensor 1	27.6B-SEN.001	Front sensor used when the conveyor is unloading pallets from Elevator B. It switches to the rear position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PE Sensor 2	28.6B-SEN.002	Rear sensor used when the conveyor is unloading pallets from Elevator B. It switches to the front position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
Motor-VSD	29.F_PCC_06B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.
Light Curtains	30. LIGTHC_B-SEN.001	Light curtains are security sensors that detect unauthorized or unexpected presence at the entrance or exit of the system.
Start/Reset Push Button	31. START_RST-PB.002	This button is used to reset the conveyors and restart their movement after the light curtain has been triggered.

6.2. Ground-Level Transfer Conveyor-Elevator A/B.

The Ground-Level Transfer Conveyors are critical components of the pallet handling system. They are responsible for changing the direction of pallet travel by approximately 90 degrees, serving both as entry and exit points between the ground-level conveyors and the elevator system.

Similar in function to the pallet chain conveyors, the transfer conveyors operate in the following modes:

- **Infeed Operation.**
- **Outfeed Operation.**
- **Dynamic Configuration.** The direction of operation (infeed or outfeed) is adjustable via the HMI, allowing flexible system behavior based on real-time needs and workflow requirements.

System Components

- **Presence Sensors (5 per conveyor):** These sensors detect the position and presence of pallets on the conveyor. Depending on the travel direction, they provide the **PLC** with real-time information about zone occupancy and readiness to accept a new pallet.
- **Table Position Sensors (2 per conveyor):** These sensors monitor the vertical position of the transfer table. They inform the **PLC** when the table is correctly positioned (raised or lowered) to align with incoming or outgoing pallets.
- **Three-Phase Conveyor Motor with Variable Speed Drive (VSD):** Powers the movement of pallets along the conveyor. The **VSD**, controlled via **PROFINET** by the **PLC**, adjusts speed and direction to ensure smooth and precise transport.

- **Three-Phase Transfer Motor with Variable Speed Drive (VSD):** Controls the raising and lowering of the transfer table. The **VSD**, also governed by **PROFINET**, ensures that the table transitions smoothly between levels to facilitate accurate pallet alignment and handover.

The following sections will provide detailed information about the specific devices installed in each transfer conveyor zone.

6.2.1. F_PCC_04A- Ground-Level Transfer Conveyor.

F_PCC_04A (Elevator A). Pallet infeed/outfeed Transfer conveyor.

Device	Tag	Description
PE Sensor 1	32.4A-SEN.001	Front sensor used when the conveyor section 1 is loading pallets onto Elevator A. It switches to the rear position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PE Sensor 2	33.4A-SEN.002	Rear sensor used when the conveyor section 1 is loading pallets onto Elevator A. It switches to the front position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PE Sensor 1	34.4AT-SEN.001	Front sensor used when the conveyor section 2 is loading pallets onto Elevator A. It switches to the rear position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PE Sensor 2	35.4AT-SEN.002	Sensor that detects the pallet at the point where the transfer takes place.
PE Sensor 3	36.4AT-SEN.003	Front sensor used when the conveyor section 2 is loading pallets onto Elevator A. It switches to the rear position when the conveyor is unloading pallets from Elevator A. Detects the presence of a pallet on the conveyor.
PX Sensor 1	37.4AT-PX-UP	This sensor detects when the table is in the raised (up) position.
PX Sensor 2	38.4AT-PX-DWN	This sensor detects when the table is in the lowered (down) position.
Motor-VSD	39.F_PCC_04A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.
Motor-VSD	40.F_PCC_04A_T	Transfer Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.2.2. F_PCC_01B- Ground-Level Transfer Conveyor.

F_PCC_01B (Elevator B). Pallet outfeed/ infeed Transfer conveyor.

Device	Tag	Description
PE Sensor 1	41.1B-SEN.001	Front sensor used when the conveyor section 1 is unloading pallets from Elevator B. It switches to the rear position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.

Device	Tag	Description
PE Sensor 2	42.1B-SEN.002	Rear sensor used when the conveyor section 1 is unloading pallets from Elevator B. It switches to the front position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PE Sensor 1	43.4BT-SEN.001	Front sensor used when the conveyor section 2 is unloading pallets from Elevator B. It switches to the rear position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PE Sensor 2	44.4BT-SEN.002	Sensor that detects the pallet at the point where the transfer takes place.
PE Sensor 3	45.4BT-SEN.003	Rear sensor used when the conveyor section 2 is unloading pallets from Elevator B. It switches to the front position when the conveyor is loading pallets onto Elevator B. Detects the presence of a pallet on the conveyor.
PX Sensor 1	46.4BT-PX-UP	This sensor detects when the table is in the raised (up) position.
PX Sensor 2	47.4BT-PX-DWN	This sensor detects when the table is in the lowered (down) position.
Motor-VSD	48.F_PCC_01B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.
Motor-VSD	49.F_PCC_04B_T	Transfer Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.3. Ground-Level Dimensioners

The ground-level dimensioners are designed to measure the dimensions of pallet loads to prevent collisions and obstructions. Their primary function is to ensure the safe and reliable operation of the system, protecting both the equipment and personnel.

System Components

- **Presence Sensors (4 units):** These sensors detect oversized pallets. If a pallet exceeds the predefined dimensional limits, it is automatically rejected by the system to prevent interference with downstream equipment.
- **Analysis Trigger Sensor:** This sensor initiates the dimensional analysis process. In combination with an encoder, it enables the system to measure the length of the pallet and determine whether it falls within acceptable parameters.
- **Encoder Length and Position Sensor:** This sensor captures the exact length and position of the pallet. It provides critical data used by the system to evaluate and validate the pallet’s compliance with dimensional constraints.

- **Barcode Reader:** The barcode reader communicates pallet entry and exit information to the WMS, enabling accurate tracking and data processing within the system.

The following sections detail the specific devices used.

6.3.1. Ground-Level Dimensioner Elevator A.

Ground-Level Dimensioner Elevator A.

Device	Tag	Description
PE Sensor 1	50. DIM1-SEN.001	Height Detection Sensor 1 detects pallets that exceed the maximum allowed height.
PE Sensor 2	51. DIM1-SEN.002	Height Detection Sensor 2 detects pallets that exceed the maximum allowed height.
PE Sensor 3	52. DIM1-SEN.003	Width Detection Sensor 1 detects pallets that exceed the maximum allowed width.
PE Sensor 4	53. DIM1-SEN.004	Width Detection Sensor 2 detects pallets that exceed the maximum allowed width.
PE Sensor 5	54. DIM1-SEN.005	This sensor initiates the dimensional analysis process
Encoder	55. DIM1-ENC	Encoder captures the length and position of the pallet as it passes through the dimensioning system
BC Scanner	56. DIM1-BCS	The barcode reader communicates pallet entry and exit information to the WMS, enabling accurate tracking and data processing within the system.

6.3.2. Ground-Level Dimensioner Elevator B.

Ground-Level Dimensioner Elevator B.

Device	Tag	Description
PE Sensor 1	57. DIM2-SEN.001	Height Detection Sensor 1 detects pallets that exceed the maximum allowed height.
PE Sensor 2	58. DIM2-SEN.002	Height Detection Sensor 2 detects pallets that exceed the maximum allowed height.
PE Sensor 3	59. DIM2-SEN.003	Width Detection Sensor 1 detects pallets that exceed the maximum allowed width.
PE Sensor 4	60. DIM2-SEN.004	Width Detection Sensor 2 detects pallets that exceed the maximum allowed width.
PE Sensor 5	61. DIM2-SEN.005	This sensor initiates the dimensional analysis process
Encoder	62. DIM2-ENC	Encoder captures the length and position of the pallet as it passes through the dimensioning system
BC Scanner	63. DIM2-BCS	The barcode reader communicates pallet entry and exit information to the WMS, enabling accurate tracking and data processing within the system.

6.4. Ground-Level Roller Door.

The ground-level roller doors are designed to help maintain the internal temperature of the storage area. They operate automatically—opening only when pallet movement is detected and closing when no activity is present.

System Components

- **Position Sensors (2 units):** These sensors monitor the status of the door and provide feedback to the PLC, indicating whether the door is in the open or closed position.
- **Actuator:** The actuator controls the movement of the roller door through two distinct signals—one for opening and one for closing—based on commands from the control system.

The following sections detail the specific devices used.

6.4.1. Ground-Level Roller Door Elevator A.

Ground-Level Roller Door Elevator A.

Device	Tag	Description
PX Sensor 1	64.RD1-PX-UP	Position Sensor – Door Open: Detects and confirms when the roller door is fully open.
PX Sensor 2	65.RD1-PX-DWN	Position Sensor – Door Closed: Detects and confirms when the roller door is fully closed.
Roller Actuator 1	66.RD1.UP.SOL.001	Actuator Signal – Open Command: Electrical signal sent to the actuator to initiate door opening.
Roller Actuator 2	67.RD1.DWN.SOL.001	Actuator Signal – Close Command: Electrical signal sent to the actuator to initiate door closing.

6.4.2. Ground-Level Roller Door Elevator B.

Ground-Level Roller Door Elevator B.

Device	Tag	Description
PX Sensor 1	68. RD2-PX-UP	Position Sensor – Door Open: Detects and confirms when the roller door is fully open.
PX Sensor 2	69.RD2-PX-DWN	Position Sensor – Door Closed: Detects and confirms when the roller door is fully closed.
Roller Actuator 1	70.RD2.UP.SOL.002	Actuator Signal – Open Command: Electrical signal sent to the actuator to initiate door opening.
Roller Actuator 2	71.RD2.DWN.SOL.002	Actuator Signal – Close Command: Electrical signal sent to the actuator to initiate door closing.

6.5. Lift Carrier Elevator A/B.

The Lift Carrier is a mechanical subsystem designed to accurately position itself within the elevator structure for secure and precise vertical pallet transport. It ensures the alignment and stability required for safe pallet loading and unloading at various levels.

System Components

- **End Stop Motors (3 units):** Three three-phase motors are responsible for accurately positioning the Shuttle stops at each level, guiding the Shuttles within the elevator. Each motor is independently controlled by a dedicated variable speed drive (VSD) to ensure smooth and precise movement during operation.
- **Locking Actuator :** These actuators receive open/close signals to lock the lift carrier securely in place once the target position is reached.
- **Magnetic Safety Actuators:** These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely (Lock/Unlock) before transport begins.
- **Actuator Position Sensors (2 units):** These sensors monitor the status of the linear motor, detecting whether they are in the extended or retracted position. This ensures proper engagement and secure positioning of the lift carrier.
- **End Stop Position Sensors (4 units):** These sensors verify the final position of the Shuttle stops by detecting alignment with the designated end stops, ensuring accurate and safe positioning for Shuttle movement.
- **Lock position Sensors (4 units):** These sensors detect the status of the locking mechanisms, verifying whether the locks are properly engaged or disengaged, which is critical for operational safety and mechanical stability.

The following sections detail the specific devices used.

6.5.1. Lift Carrier Elevator A.

Lift Carrier Elevator A.

Device	Tag	Description
Motor-VSD	72.F_END_STOP_01A	Shaft Left Variable Speed Drive (VSD) Motor: Responsible for accurately positioning the end stop.
Motor-VSD	73.F_END_STOP_02A	Shaft Right Variable Speed Drive (VSD) Motor: Responsible for accurately positioning the end stop.
Motor-VSD	74.F_END_STOP_LM_01A	Extend/Retract Variable Speed Drive (VSD) Motor: Responsible for accurately positioning the end stop.
Actuator	75.ELVA.ACT.001	Carriage Actuator. This actuator receives open/close signals to lock the lift carrier securely in place once the target position is reached.

Device	Tag	Description
Magnet Actuator	76.ELVA.MAG.001	Locking Magnet 1a. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
Magnet Actuator	77.ELVA.MAG.002	Locking Magnet 1b. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
Magnet Actuator	78.ELVA.MAG.003	Locking Magnet 2a. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
Magnet Actuator	79.ELVA.MAG.004	Locking Magnet 2b. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
PX Sensor 1	80. LMC01A-PX-RET	Carriage Retracted This sensor monitor the status of the linear end stop, detecting whether it is in the retracted position.
PX Sensor 2	81. LMC01A-PX-EXT	Carriage Extended. This sensor monitor the status of the linear end stop, detecting whether it is in the extended position.
PX Sensor 3	82. ESC01A-PX-DWN	End Stop Down Left. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
PX Sensor 4	83. ESC01B-PX-DWN	End Stop Down Right. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
PX Sensor 5	84. ESC01A-PX-SHFT	End Stop Shft Left. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
PX Sensor 6	85. ESC01B-PX-SHFT	End Stop Shft Right. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
CR-B5 Sensor 5	86. ELVA.CB-B5-L1	Lock 1 Retracted. This sensor monitors the status of the lift lock.
CR-B6 Sensor 6	87. ELVA.CB-B6-L2	Lock 2 Retracted. This sensor monitors the status of the lift lock.
CR-B7 Sensor 7	88. ELVA.CB-B7-L3	Lock 3 Retracted. This sensor monitors the status of the lift lock.
CR-B8 Sensor 8	89. ELVA.CB-B8-L4	Lock 4 Retracted. This sensor monitors the status of the lift lock.

6.5.2. Lift Carrier Elevator B.

Lift Carrier Elevator B.

Device	Tag	Description
Motor-VSD	90.F_END_STOP_01B	Shaft Left Variable Speed Drive (VSD) Motor: Responsible for accurately positioning the end stop.
Motor-VSD	91.F_END_STOP_02B	Shaft Right Variable Speed Drive (VSD) Motor: Responsible for accurately positioning the end stop.

Device	Tag	Description
Motor-VSD	92.F_END_STOP_LM_01B	Extend/Retract Variable Speed Drive (VSD) Motor: Responsible for accurately positioning the end stop.
Actuator 1	93.ELVB.ACT.001	Carriage Actuator. This actuator receives open/close signals to lock the lift carrier securely in place once the target position is reached.
Magnet Actuator	94. ELVB.MAG.001	Locking Magnet 1a. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
Magnet Actuator	95. ELVB.MAG.002	Locking Magnet 1b. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
Magnet Actuator	96. ELVB.MAG.003	Locking Magnet 2a. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
Magnet Actuator	97. ELVB.MAG.004	Locking Magnet 2b. These actuators enhance operational safety by ensuring the lift carrier engages correctly and securely before transport begins.
PX Sensor 1	98. LMC01B-PX-RET	Carriage Retracted This sensor monitor the status of the linear end stop, detecting whether it is in the retracted position.
PX Sensor 2	99.LMC01B-PX-EXT	Carriage Extended. This sensor monitor the status of the linear end stop, detecting whether it is in the extended position.
PX Sensor 3	100. ESC02A-PX-DWN	End Stop Down Left. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
PX Sensor 4	101. ESC02B-PX-DWN	End Stop Down Right. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
PX Sensor 5	102. ESC02A-PX-SHFT	End Stop Shft Left. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
PX Sensor 6	103. ESC02B-PX-SHFT	End Stop Shft Right. These sensors verify the final position of each stop by detecting its position status, ensuring accurate alignment and system reliability.
CR-B5 Sensor 5	104. ELVB.CB-B5-L1	Lock 1 Retracted. This sensor monitors the status of the lift lock.
CR-B6 Sensor 6	105. ELVB.CB-B6-L2	Lock 2 Retracted. This sensor monitors the status of the lift lock.
CR-B7 Sensor 7	106. ELVB.CB-B7-L3	Lock 3 Retracted. This sensor monitors the status of the lift lock.
CR-B8 Sensor 8	107. ELVB.CB-B8-L4	Lock 4 Retracted. This sensor monitors the status of the lift lock.

6.6. Lift Carrier Pallet Chain Conveyor Elevator A/B.

These conveyors are designed to transport pallets horizontally into or out of the lift carrier, enabling efficient loading and unloading of pallets at various levels.

Functional Capabilities:

- **Infeed Operation:** Pallets are transferred from ground level onto the lift to be transported to the designated storage levels and unloaded accordingly.
- **Outfeed Operation:** Pallets are collected from each storage level and loaded onto the lift for transport back to ground level.
- **Dynamic Configuration:** The transport direction (infeed or outfeed) can be configured via the HMI, allowing operators to adjust pallet flow based on operational requirements.

System Components:

- **Sensors (2 per conveyor):** Detect the presence and position of pallets. Depending on the configured transport direction, they inform the PLC whether the zone is occupied or available.
- **Three-phase Motor with Variable Speed Drive (VSD):** Each conveyor includes a motor controlled by a VSD, which receives commands from the PLC over PROFINET. The VSD regulates the motor's speed and direction to ensure precise and smooth pallet movement between conveyor zones.

Further details on the specific devices used in each conveyor zone are provided in the following sections.

6.6.1. F_PCC_LIFT_A- Lift Carrier Pallet Chain Conveyor Elevator A.

F_PCC_LIFT_A (Elevator A). Lift Carrier Pallet Chain Conveyor.

Device	Tag	Description
PE Sensor 1	108. LCA-SEN.001	Front sensor used when the conveyor loads pallets onto Elevator A. It activates in the rear position when the conveyor changes direction. It detects the presence of a pallet on the conveyor.
PE Sensor 2	109. LCA-SEN.002	Rear sensor used when the conveyor loads pallets onto Elevator A. It activates in the front position when the conveyor changes direction. It detects the presence of a pallet on the conveyor.
Motor-VSD	110.F_PCC_LIFT_A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.6.2. F_PCC_LIFT_B- Lift Carrier Pallet Chain Conveyor Elevator B.

F_PCC_LIFT_B (Elevator B). Lift Carrier Pallet Chain Conveyor.

Device	Tag	Description
PE Sensor 1	111. LCB-SEN.001	Front sensor used when the conveyor loads pallets onto Elevator B. It activates in the rear position when the conveyor changes direction. It detects the presence of a pallet on the conveyor.
PE Sensor 2	112. LCB-SEN.002	Rear sensor used when the conveyor loads pallets onto Elevator B. It activates in the front position when the conveyor changes direction. It detects the presence of a pallet on the conveyor.
Motor-VSD	113.F_PCC_LIFT_B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.7. Elevator A/B.

The elevator is responsible for vertically transporting pallets between the ground level and the designated storage levels. Depending on the operational configuration, it can perform both upward and downward transport, as determined by the system's flow settings.

Functional Modes:

- **Inbound Operation:** Pallets are received from ground level and transported to a designated storage level.
- **Outbound Operation:** Pallets are retrieved from a storage level and transported to ground level.
- **Dynamic Configuration:** The transport direction (inbound or outbound) can be configured via the HMI, allowing operators to adapt flow as needed.

System Components:

- **Three-Phase Motors with VSDs (2 units):** The elevator is powered by two three-phase motors, each paired with a Variable Speed Drive (VSD). The VSDs receive commands from the PLC over PROFINET to regulate motor speed, direction, and operation—ensuring smooth and controlled vertical pallet movement.
- **Elevator Chain Position Sensors (4 units):** These sensors monitor the position of the elevator chains and provide real-time status to the PLC, ensuring safe and reliable operation.
- **Lift Carrier Position Sensors – Digital (6 units):** These sensors detect the lift carrier's position at specific vertical points within the elevator shaft. They inform the PLC of the carrier's location for precise level alignment.
- **Lift Carrier Position Sensor – Analog (1 unit):** An analog sensor that continuously monitors the exact vertical position of the lift carrier. The PLC uses this data to determine when to stop or move the carrier, enabling accurate positioning at each level.

The following sections detail the specific devices used.

6.7.1. Elevator A.

Elevator A.

Device	Tag	Description
Motor-VSD	114.F_LIFT_01A	Variable Speed Drive (VSD) motor that moves the elevator.
Motor-VSD	115.F_LIFT_02A	Variable Speed Drive (VSD) motor that moves the elevator.
CR-B1 Sensor 1	116.ELVA.CB-B1-CC1	Carrier Chain 1 ok. This sensor monitors the position of the elevator chains and provide real-time status.
CR-B2 Sensor 2	117.ELVA.CB-B2-CC2	Carrier Chain 2 ok. This sensor monitors the position of the elevator chains and provide real-time status.
CR-B3 Sensor 3	118.ELVA.CB-B3-CC3	Carrier Chain 3 ok. This sensor monitors the position of the elevator chains and provide real-time status.
CR-B4 Sensor 4	119.ELVA.CB-B4-CC4	Carrier Chain 4 ok. This sensor monitors the position of the elevator chains and provide real-time status.
CR-W1 Sensor 1	120.ELVA.CB-W1-CP	Carriage Position. This sensor detects the position of the elevator conveyor at specific vertical points.
BC-B1 Sensor 1	121.ELVA.BC-B1-BP	Basic Position. This sensor detects the position of the elevator conveyor at specific vertical points (Basic Position).
BC-S1 Sensor 1	122. ELVA.BC-S1-BLS1	Bottom Limit Switch 1. This sensor detects the position of the elevator conveyor at specific vertical points
BC-S2 Sensor 2	123. ELVA.BC-S2-BLS2	Bottom Limit Switch 2. This sensor detects the position of the elevator conveyor at specific vertical points.
TC-S1 Sensor 1	124.ELVA.TC-S1-TLS1	Top Limit Switch 1. This sensor detects the position of the elevator conveyor at specific vertical points.
TC-S2 Sensor 2	125.ELVA.TC-S2-TLS2	Top Limit Switch 2. This sensor detects the position of the elevator conveyor at specific vertical points.
LT_A-Sensor	126.ELVA.LT_A-	Analog sensor that continuously monitors the exact vertical position of the lift carrier

6.7.2. Elevator B.

Elevator B.

Device	Tag	Description
Motor-VSD	127.F_LIFT_01B	Variable Speed Drive (VSD) motor that moves the elevator.
Motor-VSD	128.F_LIFT_02B	Variable Speed Drive (VSD) motor that moves the elevator.
CR-B1 Sensor 1	129. ELVB.CB-B1-CC1	Carrier Chain 1 ok. This sensor monitors the position of the elevator chains and provide real-time status.

Device	Tag	Description
CR-B2 Sensor 2	130. ELVB.CB-B2-CC2	Carrier Chain 2 ok. This sensor monitors the position of the elevator chains and provide real-time status.
CR-B3 Sensor 3	131. ELVB.CB-B3-CC3	Carrier Chain 3 ok. This sensor monitors the position of the elevator chains and provide real-time status.
CR-B4 Sensor 4	132. ELVB.CB-B4-CC4	Carrier Chain 4 ok. This sensor monitors the position of the elevator chains and provide real-time status.
CR-W1 Sensor 1	133. ELVB.CB-W1-CP	Carriage Position. This sensor detects the position of the elevator conveyor at specific vertical points.
BC-B1 Sensor 1	134. ELVB.BC-B1-BP	Basic Position. This sensor detects the position of the elevator conveyor at specific vertical points (Basic Position).
BC-S1 Sensor 1	135. ELVB.BC-S1-BLS1	Bottom Limit Switch 1. This sensor detects the position of the elevator conveyor at specific vertical points
BC-S2 Sensor 2	136. ELVB.BC-S2-BLS2	Bottom Limit Switch 2. This sensor detects the position of the elevator conveyor at specific vertical points.
TC-S1 Sensor 1	137. ELVB.TC-S1-TLS1	Top Limit Switch 1. This sensor detects the position of the elevator conveyor at specific vertical points.
TC-S2 Sensor 2	138. ELVB.TC-S2-TLS2	Top Limit Switch 2. This sensor detects the position of the elevator conveyor at specific vertical points.
LT_B-Sensor	139. ELVB.LT_B-	Analog sensor that continuously monitors the exact vertical position of the lift carrier

6.8. Level Pallet Chain Conveyors with Shuttle Access Elevator A/B.

These subsystem conveyors are located on each level of the storage system and serve as the interface between the elevator and the sorting robot. They facilitate the bi-directional transfer of pallets—either receiving pallets from the elevator for processing or sending sorted pallets back to the elevator for outbound transport. The configuration of each conveyor is controlled via the Human-Machine Interface (HMI), allowing dynamic operational adjustments.

Functional Modes:

- **Infeed Operation:** Pallets arriving from the elevator are staged in these zones at each level. The sorting robot then retrieves the pallets for storage or further handling.
- **Outfeed Operation:** Pallets are placed onto these conveyors by the sorting robot. The conveyors then transfer the pallets to the elevator for transport to ground level.
- **Dynamic Configuration:** The direction of pallet movement (infeed or outfeed) can be configured through the HMI, giving operators the flexibility to adapt to real-time operational requirements.

System Components:

- **Pallet Detection Sensors (2 per conveyor):** These sensors monitor the presence and position of pallets. Based on the configured direction, they provide real-time status updates to the PLC, indicating whether a zone is occupied or ready to receive a new pallet.
- **Three-Phase Motor with Variable Speed Drive (VSD):** Each conveyor is equipped with a three-phase motor controlled by a VSD. The VSD receives operational commands from the PLC via PROFINET, allowing for precise control of motor speed, direction, and overall movement—ensuring efficient and safe pallet transfer between the robot and the elevator.

The following sections detail the specific devices and configurations used in each conveyor zone.

6.8.1. F_PCC_WSA_01A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_01A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	140.HO-1A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	141. HO-1A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	142.F_PCC_WSA_01A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.2. F_PCC_WSA_02A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_02A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	143.HO-2A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	144 HO-2A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	145.F_PCC_WSA_02A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.3. F_PCC_WSA_03A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_03A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	146.HO-3A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	147.HO-3A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	148.F_PCC_WSA_03A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.4. F_PCC_WSA_04A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_04A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	149.HO-4A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	150.HO-4A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	151.F_PCC_WSA_04A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.5. F_PCC_WSA_05A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_05A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	152. HO-5A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	153. HO-5A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	154.F_PCC_WSA_05A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.6. F_PCC_WSA_06A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_06A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	155. HO-6A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	156. HO-6A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	157.F_PCC_WSA_06A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.7. F_PCC_WSA_07A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_07A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	158. HO-7A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	159. HO-7A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	160.F_PCC_WSA_07A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.8. F_PCC_WSA_08A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_08A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	161.HO-8A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	162.HO-8A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	163.F_PCC_WSA_08A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.9. F_PCC_WSA_09A-Level Pallet Chain Conveyors with Shuttle Access Elevator A.

F_PCC_WSA_09A (Elevator A). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	164.HO-9A-SEN.001	Front Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the rear sensor.
PE Sensor 2	165.HO-9A-SEN.002	Rear Position Sensor detects the presence of a pallet when the conveyor is set to load pallets into the area (infeed direction). When the conveyor's direction is reversed (outfeed), this sensor becomes the front sensor.
Motor-VSD	166.F_PCC_WSA_09A	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.10. F_PCC_WSA_01B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_01B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	167.HO-1B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	168.HO-1B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	169.F_PCC_WSA_01B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.11. F_PCC_WSA_02B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_02B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	170.HO-2B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	171.HO-2B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	172.F_PCC_WSA_02B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.12. F_PCC_WSA_03B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_03B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	173.HO-3B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	174.HO-3B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	175.F_PCC_WSA_03B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.13. F_PCC_WSA_04B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_04B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	176.HO-4B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	177.HO-4B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	178.F_PCC_WSA_04B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.14. F_PCC_WSA_05B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_05B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	179.HO-5B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	180.HO-5B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	181.F_PCC_WSA_05B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.15. F_PCC_WSA_06B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_06B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	182.HO-6B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	183.HO-6B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	184.F_PCC_WSA_06B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.16. F_PCC_WSA_07B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_07B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	185.HO-7B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	186.HO-7B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	187.F_PCC_WSA_07B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.17. F_PCC_WSA_08B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_08B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	188.HO-8B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	189. HO-8B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	190.F_PCC_WSA_08B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.8.18. F_PCC_WSA_09B-Level Pallet Chain Conveyors with Shuttle Access Elevator B.

F_PCC_WSA_09B (Elevator B). Level Pallet Chain Conveyors with Shuttle Access.

Device	Tag	Description
PE Sensor 1	191.HO-9B-SEN.001	The front position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the rear sensor.
PE Sensor 2	192.HO-9B-SEN.002	The rear position sensor detects the presence of a pallet when the conveyor is configured to load pallets onto the elevator (outfeed direction). When the conveyor reverses direction (infeed), this sensor becomes the front sensor.
Motor-VSD	193.F_PCC_WSA_09B	Variable Speed Drive (VSD) motor that moves the pallet according to the transport logic.

6.9. Safety System Elevator A/B.

As outlined in the system architecture, the control system utilizes an S7-1500 Safety PLC. This PLC includes dedicated safety logic and safety-rated input/output (I/O) modules. Within the PLC, the safety logic takes precedence over all other processes. If any component of the safety system—such as I/O modules or critical safety signals—fails or behaves abnormally, the PLC will immediately execute the safety logic.

Upon activation of the safety logic, the system will deactivate the main contactors, cutting power to relevant components. Using the Profinet communication protocol and its integrated Profisafe functionality, the system can issue emergency stop commands that immediately halt VSDs (Variable Speed Drives) and motors without applying a deceleration ramp, ensuring a rapid and safe stop.

The signals monitored and managed in this safety section include:

Device	Tag	Description
E-Stop	194. IN_Estop	Safety input. Local Emergency stop
E-Stop	195. MOVU_IN_Estop	Safety input. Stow Emergency stop
Door Switches 1	196. DS_01	Safety input. Front Doors OK.
Door Switches 2	197. DS_02	Safety input. Rear Doors OK.
Safety Contactor 1	198. KM010_1	Safety Output. Safety Contactor 1 KM010_1
Safety Contactor 2	199. KM010_2	Safety Output. Safety Contactor 2 KM010_2

Device	Tag	Description
Out Estop	200.MOVU_OUT_ESTOP	Safety Output. Out From PLC 1500 to Stow Emergency stop
Out Auto Mode	201.MOVU_OUT_AutoMode	Safety Output. Out From PLC 1500 to Stow Auto Mode
Out Permitier Closed	202.MOVU_OUT_PermitierClosed	Safety Output. Out From PLC 1500 to Stow Permitier Closed
General VSD ProfiSafe	203.LDrvSafe_SinaGTlg30Control.STO	ProfiSafe Output
Light Curtains A	LIGTHC_A-SEN.001	Light curtains are security sensors that detect unauthorized or unexpected presence at the entrance or exit of the system.
Light Curtains B	LIGTHC_B-SEN.001	Light curtains are security sensors that detect unauthorized or unexpected presence at the entrance or exit of the system.

7. Modes of Operation

The system's primary role is to transport pallets from ground level to each level, and from each level to ground level. Since all subsystems interact with each other, the system has three different control modes:

- Manual / Local control. This manual mode was created only for maintenance purposes where motors, actuators and other devices can be operated only for this purpose (It is recommended that this mode be used only by qualified personnel).
- Semi-Automatic / Local control This semi-automatic mode was designed to activate the system as a whole by sending and receiving pallets but only from the HMI screen where these controls will be enabled (The system will continue exchanging information with the WMS depending on whether communication continues to be established).
- Automatic / Remote control. This Automatic mode was designed to activate the system as a whole by sending and receiving pallets from the commands activated from the WMS system.
- Emergency or safe shutdown mode. This mode is the system's primary mode; when activated, its primary objective is to protect operators, products, and equipment from potential hazards.

8. Manual Mode Operation

As previously stated, manual mode is intended strictly for maintenance purposes. It is strongly recommended that only qualified and trained personnel operate the system in this mode, as it presents a high risk of injury to personnel and potential damage to equipment or mechanical components.

Operating Sequence in Manual Mode.

Permissive list		
No safety alarm should be activated. Devices that are to be run manually should not have any alarms activated.		
Step	Action	Completion criteria
1	Confirm safety contactors are energized KM010_1 and KM010_2. The system should indicate that the safety circuits engaged and operational.	System is healthy
2	Select "Manual Mode" on the main screen of the HMI.	System in manual mode
3	The operator or system user must log in, as the availability of the start button for manual operation depends on their security clearance level.	User-Login
4	Press "Start" on the main HMI screen to activate the system in manual mode.	System is running
5	Access the motor or actuator control. Navigate to the overview or maintenance screen and open the faceplate or pop-up window for the specific motor or actuator you wish to control manually.	System is running
6	Perform the required action. Operate the selected device for the maintenance task.	System is running and device required action active.

Step	Action	Completion criteria
7	Once complete, stop the device using the corresponding HMI controls.	System is running and device required action stopped.
8	Stop the system. Return to the main HMI screen and press "Stop" to fully deactivate the system and exit manual mode.	System stopped and healthy

9. Semi-Automatic Mode Operation

Semi-automatic mode is designed to activate and operate the entire pallet handling system using local conveyor controls, without intervention from the Warehouse Management System (WMS). All commands are issued and monitored directly from the HMI, where the relevant control functions are enabled. Due to the complexity of operations and the risk involved, semi-automatic mode must only be used by trained and authorized personnel. Improper use can result in serious injury or damage to equipment and mechanical systems.

The following sections will detail the execution sequence of the system in semi-automatic mode, outlining the step-by-step operational flow and the individual sequences of each subsystem involved. This includes the handling of pallets across conveyors, elevators, transfer zones, and associated automation components, all controlled locally via the HMI.

Mode activation sequence.

Permissive list		
No safety alarm should be activated. Devices that are to be run Semi-Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	Confirm safety contactors are energized KM010_1 and KM010_2. The system should indicate that the safety circuits are engaged and operational.	System is healthy
2	Select "Semiautomatic Mode" on the main screen of the HMI.	System in Semiautomatic mode
3	The system operator or user must be logged in, as the availability of the start button for semi-automatic operation depends on their security clearance level.	User-Login
4	Press "Start" on the main HMI screen to activate the system in Semiautomatic Mode.	System is running

9.1. Semi-Automatic Infeed Ground Level Conveyor Sequence Elevator A

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Semi-Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	Ensure the conveyor is completely stopped before placing the pallet to prevent any operational risk.	F_PCC_01A and F_PCC_02A Stopped.
2	When the operator is loading a pallet, the light curtain is triggered to stop the conveyors, ensuring the system halts in case it begins operating automatically.	LIGTHC_A-SEN.001 Alarm Activated (On)
3	The operator manually loads the pallet onto the induction conveyor.	F_PCC_01A and F_PCC_02A Stopped.
4	From the HMI, on the INFEED/OUTFEED screen, start the conveyor by selecting "Start" in the Elve A Control Input box. The operator must also press the button to reset (START_RST-PB.001) the alarm, clear the light curtain status, and initiate the start sequence.	F_PCC_01A and F_PCC_02A Running.
5	As the conveyor starts, the sensor on Dimensioner A is triggered, beginning pallet size evaluation and also the scanning of the barcode.	Dimensioner A is Testing and running .
6	The evaluation ends when the DIM1-SEN.005 no longer detects the pallet, when any sizing sensor detects an oversize condition, or when a barcode reading error occurs.	Dimensioner A Stopped.
7	Once dimensioning completes, the barcode scanner is triggered and sends the pallet data to the WMS, which returns a location assignment.	WMS Response
8	If the result is negative (either from dimensioning or WMS response): the Infeed/Outfeed control stops. The operator is then given the option to return the pallet by pressing the Start/reset button (START_RST-PB.001), which changes the direction of the conveyor and allows the pallet to be removed.	Dimensioner A, F_PCC_01A and F_PCC_02A Running.

Step	Action	Completion criteria
9	Once the pallet is positioned for unloading—either for dimension adjustments or replacement with a pallet bearing a valid barcode—it is necessary to acknowledge the alarm on the Infeed/Outfeed HMI screen. After that, the Start/Reset button (START_RST-PB.001) must be pressed again to allow the pallet to be re-evaluated by the dimensioner.	Dimensioner A, F_PCC_01A and F_PCC_02A Stopped.
10	If the result is positive, the pallet continues until it activates sensor 2A-SEN.001, stopping conveyors F_PCC_01A and F_PCC_02A, and starting F_PCC_03A conveyor.	F_PCC_01A and F_PCC_02A Stopped. F_PCC_03A running.
11	The operator opens ROLLER DOOR A from the HMI. When RD1-PX-UP (door open feedback) is triggered, the pallet continues moving.	F_PCC_02A and F_PCC_03A running.
12	After the pallet completely passes through the roller door, the operator closes it via RD1.DWN.SOL.001 on the HMI. Confirmation comes from the door closed feedback signal.	RD1-PX-DWN Activated (On)
13	When the pallet reaches 3A-SEN.001, the transfer conveyor table raises.	F_PCC_03A Stopped. F_PCC_04A_T running.
14	Upon feedback signal 4AT-PX-UP (Table Up), the table lift motor (F_PCC_04A_T) stops and the pallet conveyor resumes.	F_PCC_04A_T Stopped. F_PCC_03A and F_PCC_04A running.
15	When the pallet activates sensor 4AT-SEN.002, the table lowers via its motor	F_PCC_04A_T running.
16	When the table down feedback (4AT-PX-DWN). is received: The transfer table motor and previous conveyors stop, and the last infeed conveyor (F_PCC_05A) is activated.	F_PCC_03A, F_PCC_04A and F_PCC_04A_T Stopped. F_PCC_05A running.
17	When 5A-SEN.001 detects pallet presence, the conveyor stops and informs Elevator A that a pallet is ready.	F_PCC_05A Stopped.
18	If the WMS has confirmed pallet acceptance and Elevator A is at Level 1, Ready to load and healthy, the conveyors restart, and the pallet is loaded into the elevator.	F_PCC_05A and F_PCC_LIFT_A running.

Step	Action	Completion criteria
19	The LCA-SEN.001 sensor detects the pallet on the lift conveyor. This conveyor stops, informing the elevator it is ready for vertical transport.	F_PCC_05A and F_PCC_LIFT_A Stopped.
20	The system returns to the ready state for the next pallet cycle.	

9.2. Semi-Automatic Infeed-Elevator A Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Semi-Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	Before starting the elevator movement, the operator must check first if the chain indicators are in OK status.	ELVA.CB-B1-CC1, ELVA.CB-B2-CC2, ELVA.CB-B3-CC3 and ELVA.CB-B4-CC4 Activated (On)
2	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVA.CB-B5-L1, ELVA.CB-B6-L2, ELVA.CB-B7-L3 and ELVA.CB-B8-L4 Activated (On)
3	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2) and bottom (ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1) of the elevator.	ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2 Activated (On). ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1 Deactivated (Off).
4	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 01 STATUS" section, press the button to enable the elevator.	Enable status running.
5	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01A-PX-RET, ESC01A-PX-SHFT and ESC01B-PX-SHFT Activated (On).

Step	Action	Completion criteria
6	If the elevator is on another level, the operator must send the elevator to level 1. This is done from the HMI's ELEVATORS screen. After the operator assigns the level, the elevator motors start running.	F_LIFT_01A and F_LIFT_02A Running.
7	When the level sensor ELVA.LT_A-reaches the target and the level sensor 1 ELVA.BC-B1-BP is activated, the motors are turned off, stopping the elevator.	F_LIFT_01A and F_LIFT_02A Stopped.
8	If the WMS has confirmed pallet acceptance and Elevator A is at Level 1, Ready to load and healthy, the conveyors restart, and the pallet is loaded into the elevator.	F_PCC_05A and F_PCC_LIFT_A running.
9	The LCA-SEN.001 sensor detects the pallet on the lift conveyor. This conveyor stops, informing the elevator it is ready for vertical transport.	F_PCC_05A and F_PCC_LIFT_A Stopped.
10	Once the pallet is inside the lift, the operator must select the level to which the pallet will be transported on the HMI's ELEVATORS screen.	Level assigned by the operator
11	With the "End Stop" motors stopped and the position sensors activated, the elevator is ready to move and turns on the elevator motors, transporting the pallet to the target level.	F_LIFT_01A and F_LIFT_02A Running.
12	When the elevator level sensor (ELVA.LT_A-) detects that it has reached the target level and the lift carrier sensor (ELVA.CB-W1-CP) is activated, the elevator motors stop and the elevator carrier motors and the level it reached are activated to unload the pallet (level 4 as an example).	F_LIFT_01A and F_LIFT_02A Stopped. F_PCC_LIFT_A and F_PCC_WSA_04A Running.
13	Once the pallet is loaded on the level conveyor (Example level 4) and is detected by the front sensor (HO-1A-SEN.001) the conveyor motor will stop and inform the WMS and .Stow system that the area is occupied and is ready to be unloaded by the robot.	F_PCC_WSA_04A Stopped. Signal to Stow and WMS Activated.
14	When the robot picks up the package, the area sensor is released or deactivated and with this the area is free to receive the next pallet	HO-4A-SEN.001 and HO-4A-SEN.002 Deactivated (Off).

9.3. Semi-Automatic Outfeed-Elevator B Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Semi-Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	Before starting the elevator movement, the operator must check first if the chain indicators are in OK status.	ELVB.CB-B1-CC1, ELVB.CB-B2-CC2, ELVB.CB-B3-CC3 and ELVB.CB-B4-CC4 Activated (On)
2	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVB.CB-B5-L1, ELVB.CB-B6-L2, ELVB.CB-B7-L3 and ELVB.CB-B8-L4 Activated (On)
3	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVB.TC-S1-TLS1 and ELVB.TC-S2-TLS2) and bottom (ELVB.BC-S1-BLS1 and ELVB.BC-S2-BLS2) of the elevator.	ELVB.BC-S1-BLS1 and ELVB.TC-S2-TLS2 Activated (On). ELVB.BC-S2-BLS2 and ELVB.TC-S1-TLS1 Deactivated (Off).
4	When the robot loads the pallet in the area or level of the conveyor that the elevator loads (Example level 4), this pallet activates the sensors informing the system that this area is occupied and the Stow and WMS system informs the PLC that this conveyor is loaded to send the pallet to the ground level.	HO-4B-SEN.001 Activated (On).
5	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 02 STATUS" section, press the button to enable the elevator.	Enable status running.
6	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01B-PX-RET, ESC02A-PX-SHFT and ESC02B-PX-SHFT Activated (On).

Step	Action	Completion criteria
7	If the elevator is on a different level, the operator must direct the elevator to the level where the pallet will be loaded (e.g., level 4). This is done from the HMI's ELEVATORS screen. When the operator selects the level, the elevator turns on its motors.	F_LIFT_01B and F_LIFT_02B Running.
8	When the elevator level sensor (ELVB.LT_B-) detects that it has reached the target level (level 4 as an example) and the elevator carrier sensor (ELVB.CB-W1-CP) is activated, the elevator motors stop and the elevator carrier conveyor motor and the level reached conveyor motor are activated to load the pallet onto the elevator.	F_LIFT_01B and F_LIFT_02B Stopped. F_PCC_WSA_04B and F_PCC_LIFT_B Running.
9	When the LCB-SEN.001 sensor of the lift carrier is activated, it stops the conveyor motors and informs the system that the pallet is inside the elevator and ready to be transported.	F_PCC_WSA_04B and F_PCC_LIFT_B Stopped.
10	Once the pallet is inside the elevator, the operator must send it to level 1, or ground level. This action is executed on the HMI's ELEVATORS screen. Once the level is selected, the elevator motors will begin running.	F_LIFT_01B and F_LIFT_02B Running.
11	When the level sensor reaches its target (ELVB.LT_B-) and the elevator base sensor (ELVB.BC-B1-BP) is activated, the elevator motors stop.	F_LIFT_01B and F_LIFT_02B Stopped.
12	When the lift carrier is in position to be unloaded and the ground level Outfeed control is running (Actuated by the operator from the HMI on the INFEED/OUTFEED screen), the table transfer motor runs to lower the table.	F_PCC_04B_T Running.
13	When the transfer table is down, activating the position sensor 4BT-PX-DWN, the motor stops.	F_PCC_04B_T Stopped
14	When the transfer table is down and the motor is stopped, the lift carrier F_PCC_LIFT_B and conveyors begin to move, unloading the pallet from the elevator.	F_PCC_LIFT_B and F_PCC_01B Running
15	When the 1B-SEN.001 sensor of the conveyor and sensor 4BT-SEN.002 are activated, the transfer table runs the F_PCC_04B_T motor to raise the table.	F_PCC_04B_T Running
16	When the table rises, confirmed by the sensor 4BT-PX-UP, the motors of the transfer conveyor and the lift carrier stop, ending the process in the elevator and continuing the transport on the ground level conveyors.	F_PCC_04B_T, F_PCC_LIFT_B and F_PCC_01B Stopped

9.4. Semi-Automatic Outfeed Ground Level Conveyor Sequence Elevator B

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Semi-Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	Before starting, the operator must verify that the conveyor has enough space to receive the pallet. To do this, it is suggested to free up space by removing the pallet from the conveyor when it is stopped to prevent operational risks.	F_PCC_05B and F_PCC_06B Stopped.
2	As mentioned in the elevator sequence when the elevator is in the unloading position, from the HMI, under the INFEED/OUTFEED screen, initiate the conveyor by selecting "Start" in the Control Elve B Outfeed box.	Outfeed control is running
3	When the control is started, the first thing the system does is determine the position of the transfer table and lower it by running the motor.	F_PCC_04B_T Running.
4	When the transfer table is down, activating the position sensor 4BT-PX-DWN, the motor stops.	F_PCC_04B_T Stopped
5	When the transfer table is down and the motor is stopped, the lift carrier F_PCC_LIFT_B and F_PCC_01B conveyors begin to move, unloading the pallet from the elevator.	F_PCC_LIFT_B and F_PCC_01B Running
6	When the 1B-SEN.001 sensor of the conveyor and sensor 4BT-SEN.002 are activated, the transfer table runs the F_PCC_04B_T motor to raise the table.	F_PCC_04B_T Running
7	When the table rises, confirmed by the sensor 4BT-PX-UP, the motors of the transfer conveyor and the lift carrier stop, ending the process in the elevator and continuing the transport on the ground level turning on the next conveyor.	F_PCC_04B_T, F_PCC_01B and F_PCC_LIFT_B Stopped. F_PCC_02B Running
8	When the 2B-SEN.001 sensor of this conveyor is activated, the next zone or conveyor turns on to continue transporting the pallet.	F_PCC_02B, F_PCC_03B Running

Step	Action	Completion criteria
9	When the 3B-SEN.001sensor of this conveyor is activated by the pallet, this conveyor and the previous one stop and the next zone or conveyor turns on.	F_PCC_02B, F_PCC_03B Stopped. F_PCC_04B Running
10	The operator opens ROLLER DOOR B **(**RD2.UP.SOL.002**)** from the HMI. When RD2-PX-UP (door open feedback) is triggered, the pallet continues moving.	F_PCC_03B and F_PCC_04B Running.
11	After the pallet completely passes through the roller door, the operator closes it via RD2.DWN.SOL.002 on the HMI. Confirmation comes from the door closed feedback signal.	RD2-PX-DWN Activated (On)
12	When the 4B-SEN.001 sensor of this conveyor is activated, the next two zones or conveyors are turned on to continue transporting the pallet and the previous zone stops.	F_PCC_03B Stopped. F_PCC_04B, F_PCC_05B and F_PCC_06B Running
13	Finally, when the last conveyor sensor is activated 6B-SEN.001, it stops the conveyors, and the operator can stop the control from the "INFEED OUTFEED" screen of the HMI. Locally, it is ensured that the conveyors are not moving to reduce risks and then remove the pallet.	F_PCC_04B, F_PCC_05B and F_PCC_06B Stopped.
14	The system returns to the ready state for the next pallet cycle.	

10. Automatic Mode Operation

Automatic mode is designed to activate and operate a large portion of the pallet handling system using the conveyor's remote controls, with intervention from the Warehouse Management System (WMS). All commands are issued from the WMS and monitored directly from the HMI and WMS, where the relevant control functions are enabled. Due to the complexity of the operations and the risk involved, Automatic mode should only be used and monitored by trained and authorized personnel. Improper use can result in serious injury or damage to equipment and mechanical systems.

The following sections will detail the system's execution sequence in Automatic mode, describing the step-by-step operational flow and the individual sequences for each subsystem involved. This includes pallet handling through conveyors, elevators, transfer zones, and associated automation components, all controlled locally through the HMI.

Mode activation sequence.

Permissive list		
No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	Confirm safety contactors are energized KM010_1 and KM010_2. The system should indicate that the safety circuits are engaged and operational.	System is healthy
2	Select "Automatic Mode" on the main screen of the HMI.	System in Automatic mode
3	Press "Start" on the main HMI screen to activate the system in Automatic Mode.	System is running

10.1. Automatic Infeed Ground Level Conveyor Sequence Elevator A

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	Ensure the conveyor is completely stopped before placing the pallet to prevent any operational risk.	F_PCC_01A and F_PCC_02A Stopped.
2	When the operator is loading a pallet, the light curtain is triggered to stop the conveyors, ensuring the system halts in case it begins operating automatically.	LIGTHC_A-SEN.001 Alarm Activated (On)
3	The operator manually loads the pallet onto the induction conveyor.	F_PCC_01A and F_PCC_02A Stopped.
4	From the HMI, under the INFEED/OUTFEED screen, initiate the conveyor by selecting "Start" in the Control Elve A Infeed box if not already active. The operator must also press the button to reset (START_RST-PB.001) the alarm, clear the light curtain status, and initiate the start sequence.	F_PCC_01A and F_PCC_02A Running.
5	As the conveyor starts, the sensor DIM1-SEN.005 on Dimensioner A is triggered, beginning pallet size evaluation and also the scanning of the barcode.	Dimensioner A is Testing and running.
6	The evaluation ends when the DIM1-SEN.005 no longer detects the pallet, when any dimensioner sensor detects an oversize condition. The barcode is evaluated after the pallet completes the pass through the dimesnioner.	Dimensioner A Stopped.
7	Once dimensioning completes, the barcode scanner is triggered and sends the pallet data to the WMS, which returns a location assignment.	WMS Response
8	If the result is negative (either from dimensioning or WMS response): the Infeed/Outfeed control stops. The operator is then given the option to return the pallet by pressing the Start/reset button (START_RST-PB.001), which changes the direction of the conveyor and allows the pallet to be removed.	Dimensioner A, F_PCC_01A and F_PCC_02A Running.

Step	Action	Completion criteria
9	Once the pallet is positioned for unloading—either for dimension adjustments or replacement with a pallet bearing a valid barcode—it is necessary to acknowledge the alarm on the Infeed/Outfeed HMI screen. After that, the Start/Reset button (START_RST-PB.001) must be pressed again to allow the pallet to be re-evaluated by the dimensioner.	Dimensioner A, F_PCC_01A and F_PCC_02A Stopped.
10	If the result is positive, the pallet continues until it activates sensor 2A-SEN.001, stopping conveyors F_PCC_01A and F_PCC_02A, and starting F_PCC_03A conveyor. This is reported to the WMS to assign the task to the elevator.	F_PCC_01A and F_PCC_02A Stopped. F_PCC_03A running.
11	The system opens theROLLER DOOR A automatically when it is detected by the storage area's entry sensor. When RD1-PX-UP (door open feedback) is triggered, the pallet continues moving.	F_PCC_02A and F_PCC_03A running.
12	After the pallet completely passes through the roller door the system automatically closes roller door RD1.DWN.SOL.001. The closed-door feedback signal confirms the operation.	RD1-PX-DWN Activated (On)
13	When the pallet reaches 3A-SEN.001, the transfer conveyor table raises.	F_PCC_03A Stopped. F_PCC_04A_T running.
14	Upon feedback signal 4AT-PX-UP (Table Up), the table lift motor (F_PCC_04A_T) stops and the pallet conveyor resumes.	F_PCC_04A_T Stopped. F_PCC_03A and F_PCC_04A running.
15	When the pallet activates sensor 4AT-SEN.002, the table lowers via its motor	F_PCC_04A_T running.
16	When the table down feedback (4AT-PX-DWN). is received: The transfer table motor and previous conveyors stop, and the last infeed conveyor (F_PCC_05A) is activated.	F_PCC_03A, F_PCC_04A and F_PCC_04A_T Stopped. F_PCC_05A running.
17	When 5A-SEN.001 detects pallet presence, the conveyor stops and informs the WMS and the Elevator A that a pallet is ready.	F_PCC_05A Stopped.
18	If the WMS confirms pallet acceptance and assigns the task to Elevator A—provided it is at Level 1, in a ready state, and operating normally—the conveyors will restart, and the pallet will be loaded into the elevator.	F_PCC_05A and F_PCC_LIFT_A running.

Step	Action	Completion criteria
19	The LCA-SEN.001 sensor detects the pallet on the lift conveyor. This conveyor stops, informing the elevator it is ready for vertical transport.	F_PCC_05A and F_PCC_LIFT_A Stopped.
20	The system returns to the ready state for the next pallet cycle.	

10.2. Automatic Infeed-Elevator A Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	Before starting the elevator movement for the first time each day, the operator must check first if the chain indicators are in OK status.	ELVA.CB-B1-CC1, ELVA.CB-B2-CC2, ELVA.CB-B3-CC3 and ELVA.CB-B4-CC4 Activated (On)
2	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVA.CB-B5-L1, ELVA.CB-B6-L2, ELVA.CB-B7-L3 and ELVA.CB-B8-L4 Activated (On)
3	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2) and bottom (ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1) of the elevator.	ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2 Activated (On). ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1 Deactivated (Off).
4	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 01 STATUS" section, press the button to enable the elevator.	Enable status running.

Step	Action	Completion criteria
5	The elevator can perform several types of movements, each assigned by the WMS system based on priority. These movement purposes include Pallet Pickup, Pallet Dropoff, Shuttle Pickup, Shuttle Dropoff, and no reason movements. In this example, the ground-level transporter scanned the pallet's barcode and sent the information to the WMS system, which then assigned the elevator the task of Pallet Pickup.	The WMS system assigns the Pallet Pickup task
6	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the "End Stop" motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01A-PX-RET, ESC01A-PX-SHFT and ESC01B-PX-SHFT Activated (On).
7	If the elevator is on another level, the WMS system will send the order to the elevator to go to level 1. Once the level is assigned, the elevator motors are activated.	F_LIFT_01A and F_LIFT_02A Running.
8	When the level sensor ELVA.LT_A-reaches the target and the level sensor 1 ELVA.BC-B1-BP is activated, the motors are turned off, stopping the elevator.	F_LIFT_01A and F_LIFT_02A Stopped.
9	If the WMS has confirmed pallet acceptance and Elevator A is at Level 1, Ready to load and healthy, the conveyors restart, and the pallet is loaded into the elevator.	F_PCC_05A and F_PCC_LIFT_A running.
10	The LCA-SEN.001 sensor detects the pallet on the lift conveyor. This conveyor stops, informing the and the WMS system that it is ready for vertical transport.	F_PCC_05A and F_PCC_LIFT_A Stopped.
11	Once the pallet is inside the elevator and the WMS has provided the destination level, this information is assigned to the elevator to execute the transport operation.	The WMS system assigns the Pallet Dropoff task.
12	With the "End Stop" motors stopped and the position sensors activated, the elevator is ready to move and turns on the elevator motors, transporting the pallet to the target level.	F_LIFT_01A and F_LIFT_02A Running.
13	When the elevator level sensor (ELVA.LT_A-) detects that it has reached the target level and the lift carrier sensor (ELVA.CB-W1-CP) is activated, the elevator motors stop and the elevator carrier motors and the level it reached are activated to unload the pallet (level 4 as an example).	F_LIFT_01A and F_LIFT_02A Stopped. F_PCC_LIFT_A and F_PCC_WSA_04A Running.

Step	Action	Completion criteria
14	Once the pallet is loaded on the level conveyor (Example level 4) and is detected by the front sensor (HO-4A-SEN.001) the conveyor motor will stop and inform Stow system and WMS system that the area is occupied and is ready to be unloaded by the robot.	F_PCC_WSA_04A Stopped. Signal to Stow and WMS Activated.
15	When the robot picks up the package, the area sensor is deactivated, indicating that the area is now free to receive the next pallet.	HO-4A-SEN.001 and HO-4A-SEN.002 Deactivated (Off).

10.3. Automatic Outfeed-Elevator B Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	Before starting the elevator for the first time each day, the operator must check first if the chain indicators are in OK status.	ELVB.CB-B1-CC1, ELVB.CB-B2-CC2, ELVB.CB-B3-CC3 and ELVB.CB-B4-CC4 Activated (On)
2	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVB.CB-B5-L1, ELVB.CB-B6-L2, ELVB.CB-B7-L3 and ELVB.CB-B8-L4 Activated (On)
3	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVB.TC-S1-TLS1 and ELVB.TC-S2-TLS2) and bottom (ELVB.BC-S1-BLS1 and ELVB.BC-S2-BLS2) of the elevator.	ELVB.BC-S1-BLS1 and ELVB.TC-S2-TLS2 Activated (On). ELVB.BC-S2-BLS2 and ELVB.TC-S1-TLS1 Deactivated (Off).

Step	Action	Completion criteria
4	When the robot places a pallet on the conveyor at the level that feeds the elevator (e.g., Level 4), it triggers sensors that notify the system that the area is occupied. Based on this input, the WMS assigns the elevator the task of transporting the pallet to the ground level.	HO-4B-SEN.001 Activated (On).
5	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 02 STATUS" section, press the button to enable the elevator.	Enable status running.
6	The elevator can perform several types of movements, each assigned by the WMS system based on priority. These movement purposes include Pallet Pickup, Pallet Dropoff, Shuttle Pickup, and Shuttle Dropoff. In this example, the WMS system has the information of the Pallet to be extracted and assigns the Pallet Pickup task from level 4 to the elevator.	The WMS system assigns the Pallet Pickup task
7	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01B-PX-RET, ESC02A-PX-SHFT and ESC02B-PX-SHFT Activated (On).
8	If the lift is on a different level, the WMS system will send a command to move the lift to the level where the pallet will be loaded (e.g., level 4). When the level is selected, the lift activates its motors.	F_LIFT_01B and F_LIFT_02B Running.
9	When the elevator level sensor (ELVB.LT_B-) detects that it has reached the target level (level 4 as an example) and the elevator carrier sensor (ELVB.CB-W1-CP) is activated, the elevator motors stop and the elevator carrier conveyor motor and the level reached conveyor motor are activated to load the pallet onto the elevator.	F_LIFT_01B and F_LIFT_02B Stopped. F_PCC_WSA_04B and F_PCC_LIFT_B Running.
10	When the LCB-SEN.001 sensor of the lift carrier is activated, it stops the conveyor motors and informs the WMS system that the pallet is inside the elevator and ready to be transported.	F_PCC_WSA_04B and F_PCC_LIFT_B Stopped.
11	Once the pallet is inside the elevator and the WMS has provided the destination level, this information is assigned to the elevator to execute the transport operation.	The WMS system assigns the Pallet Dropoff task.

Step	Action	Completion criteria
12	Following the assignment from the WMS system and selection of the level, the elevator motors are activated.	F_LIFT_01B and F_LIFT_02B Running.
13	When the level sensor reaches its target (ELVB.LT_B-) and the elevator base sensor (ELVB.BC-B1-BP) is activated, the elevator motors stop.	F_LIFT_01B and F_LIFT_02B Stopped.
14	When the lift carrier is in position to be unloaded and the ground level Outfeed control is running (Actuated by the operator from the HMI on the INFEED/OUTFEED screen), the table transfer motor runs to lower the table.	F_PCC_04B_T Running.
15	When the transfer table is down, activating the position sensor 4BT-PX-DWN, the motor stops.	F_PCC_04B_T Stopped
16	When the transfer table is down and the motor is stopped, the lift carrier F_PCC_LIFT_B and conveyors begin to move, unloading the pallet from the elevator.	F_PCC_LIFT_B and F_PCC_01B Running
17	When the 1B-SEN.001 sensor of the conveyor and sensor 4BT-SEN.002 are activated, the transfer table runs the F_PCC_04B_T motor to raise the table.	F_PCC_04B_T Running
18	When the table rises, confirmed by the sensor 4BT-PX-UP, the motors of the transfer conveyor and the lift carrier stop, ending the process in the elevator and continuing the transport on the ground level conveyors.	F_PCC_04B_T, F_PCC_LIFT_B and F_PCC_01B Stopped

10.4. Automatic Outfeed Ground Level Conveyor Sequence Elevator B

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	Before starting, the operator must verify that the conveyor has enough space to receive the pallet. To do this, it is suggested to free up space by removing the pallet from the conveyor when it is stopped to prevent operational risks.	F_PCC_05B and F_PCC_06B Stopped.

Step	Action	Completion criteria
2	When the operator is unloading a pallet, the light curtain alarm is activated to stop the conveyors, ensuring that the system will automatically stop.	LIGTHC_B-SEN.001 Activated (On)
3	As mentioned in the elevator sequence, when the elevator is in the unload position, the operator must press the button (LIGTHC_B-SEN.001) to reset the alarm, clear the light curtain status and initiate the start sequence.	Outfeed control is running
4	When the control is started, the first thing the system does is determine the position of the transfer table and lower it by running the motor.	F_PCC_04B_T Running.
5	When the transfer table is down, activating the position sensor 4BT-PX-DWN, the motor stops.	F_PCC_04B_T Stopped
6	When the transfer table is down and the motor is stopped, the lift carrier F_PCC_LIFT_B and F_PCC_01B conveyors begin to move, unloading the pallet from the elevator.	F_PCC_LIFT_B and F_PCC_01B Running
7	When the 1B-SEN.001 sensor of the conveyor and sensor 4BT-SEN.002 are activated, the transfer table runs the F_PCC_04B_T motor to raise the table.	F_PCC_04B_T Running
8	When the 2B-SEN.001 sensor of this conveyor is activated, the next zone or conveyor turns on to continue transporting the pallet.	F_PCC_02B, F_PCC_03B Running
9	When the 3B-SEN.001sensor of this conveyor is activated by the pallet, this conveyor and the previous one stop and the next zone or conveyor turns on.	F_PCC_02B, F_PCC_03B Stopped. F_PCC_04B Running
10	The system opens ROLLER DOOR B **(**RD2.UP.SOL.002) when the pallet reaches its current position. When RD2-PX-UP (roller door open feedback) is triggered, the pallet continues moving.	F_PCC_03B and F_PCC_04B Running.
11	Once the pallet has completely passed through the roller door, the system closes the door rollers RD2.DWN.SOL.002. Confirmation comes from the door closed feedback signal.	RD2-PX-DWN Activated (On)
12	At this point, the scanner reads the pallet's barcode, informing the WMS system of the pallet that came out.	send the barcode to the WMS

Step	Action	Completion criteria
13	When the 4B-SEN.001 sensor of this conveyor is activated, the next two zones or conveyors are turned on to continue transporting the pallet and the previous zone stops.	F_PCC_03B Stopped. F_PCC_04B, F_PCC_05B and F_PCC_06B Running
14	Finally, when the last conveyor sensor is activated 6B-SEN.001, it stops the conveyors.	F_PCC_04B, F_PCC_05B and F_PCC_06B Stopped.
15	Upon removing the pallet, the operator will trigger the light curtain, stopping the conveyor completely. Once the pallet is removed, the operator must reset the alarm from the local button, ending the cycle so that new pallets arriving at this point can be transported.	LIGTHC_B-SEN.001 Alarm deactivated (Off)

11. Automatic Mode Redundant Elevator Operation

The previous section outlined the standard automatic operating configuration, in which:

- Elevator A handles pallet infeed (loading pallets into storage from the ground level).
- Elevator B handles pallet outfeed (removing pallets from storage to the ground level).

However, the system supports a redundancy feature, allowing the direction of pallet flow for either elevator to be modified. This enables one elevator to perform both infeed and outfeed operations, increasing system flexibility and reliability during maintenance or operational adjustments.

Flow Direction Change Execution

- The flow direction for each elevator can be reconfigured via the general HMI screen (ELEVATOR).
- The operator selects the desired operating mode (infeed or outfeed) for each elevator.

Flow Direction Change Conditions

To ensure safe and uninterrupted operations, the following rules apply when switching an elevator's flow direction:

1. Changing from Infeed to Outfeed:

- Any pallet currently being inducted (infeed sequence) must complete the cycle before the system allows the elevator to switch to outfeed.
- While switching, the system blocks new pallet infeed commands until the transition is complete.

2. Changing from Outfeed to Infeed:

- Any pallet currently being extracted (outfeed sequence) must complete its unloading before allowing the elevator to switch to infeed.
- During this change, no new outfeed orders are accepted for that elevator.

The following sections describe the automated operating sequences that occur when the system is configured in redundant mode—where either Elevator A or Elevator B can dynamically perform both infeed and outfeed tasks based on the selected configuration in the HMI (ELEVATOR screen).

These sequences ensure seamless integration of pallet movement, avoiding operational conflicts and maintaining system safety and efficiency. The logic automatically handles direction changes, pallet flow control, and the synchronization of conveyors, dimensioners, scanners, and lifts based on the active flow direction.

Each sequence is executed in real time according to system status, pallet position, and feedback from devices such as sensors, actuators, and VSD-controlled motors.

As a first step, if the system is not running, it is necessary to select automatic mode and start the system.

Mode activation sequence.

Permissive list		
No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	Confirm safety contactors are energized KM010_1 and KM010_2. The system should indicate that the safety circuits are engaged and operational.	System is healthy
2	Select "Automatic Mode" on the main screen of the HMI.	System in Automatic mode
3	Press "Start" on the main HMI screen to activate the system in Automatic Mode.	System is running

11.1. Automatic Outfeed-Elevator A Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	If the elevator is running or executing a task and is transporting pallets, skip to step 7. If it is not executing sequences, continue with step 2.	
2	Before starting the elevator movement, the operator must check first if the chain indicators are in OK status.	ELVA.CB-B1-CC1, ELVA.CB-B2-CC2, ELVA.CB-B3-CC3 and ELVA.CB-B4-CC4 Activated (On)
3	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVA.CB-B5-L1, ELVA.CB-B6-L2, ELVA.CB-B7-L3 and ELVA.CB-B8-L4 Activated (On)
4	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2) and bottom (ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1) of the elevator.	ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2 Activated (On). ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1 Deactivated (Off).
5	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 01 STATUS" section, press the button to enable the elevator.	Enable status running.
6	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01A-PX-RET, ESC01A-PX-SHFT and ESC01B-PX-SHFT Activated (On).

Step	Action	Completion criteria
7	To change the elevator's operation to redundant mode or flow change, the operator must enter the ELEVATORS screen on the HMI and select the flow direction from the Infeed to Outfeed button.	Elevator and WMS in redundant outfeed mode on request
8	If the lift is in operation or running a sequence and is transporting pallets, the WMS system and the lift must first complete the existing task before changing the direction of pallet flow.	Elevator and WMS in redundant outfeed mode on request
9	When the PLC and WMS evaluate that there are no more pallets or tasks to transport to the storage area, the flow is changed in the elevator.	Elevator in redundant outfeed mode running
10	The elevator can perform several types of movements, each assigned by the WMS system based on priority. These movement purposes include Pallet Pickup, Pallet Dropoff, Shuttle Pickup, and Shuttle Dropoff. In this example, the WMS system has the information of the Pallet to be extracted and assigns the Pallet Pickup task from level 4 to the elevator.	The WMS system assigns the Pallet Pickup task
11	When the robot places a pallet on the conveyor at the level that feeds the elevator (e.g., Level 4), it triggers sensors that notify the system that the area is occupied. Based on this input, the WMS assigns the elevator the task of transporting the pallet to the ground level.	HO-4B-SEN.001 or HO-4B-SEN.002 Activated (On).
12	If the lift is on a different level, the WMS system will send a command to move the lift to the level where the pallet will be loaded (e.g., level 4). When the level is selected, the lift activates its motors.	F_LIFT_01A and F_LIFT_02A Running.
13	When the elevator level sensor (ELVA.LT_A-) detects that it has reached the target level (level 4 as an example) and the elevator carrier sensor (ELVA.CB-W1-CP) is activated, the elevator motors stop and the elevator carrier conveyor motor and the level reached conveyor motor are activated to load the pallet onto the elevator.	F_LIFT_01A and F_LIFT_02A Stopped. F_PCC_WSA_04A and F_PCC_LIFT_A Running.
14	When the LCA-SEN.002sensor of the lift carrier is activated, it stops the conveyor motors and informs the WMS system that the pallet is inside the elevator and ready to be transported.	F_PCC_WSA_04A and F_PCC_LIFT_A Stopped.

Step	Action	Completion criteria
15	Once the pallet is inside the elevator and the WMS has provided the destination level, this information is assigned to the elevator to execute the transport operation.	The WMS system assigns the Pallet Dropoff task.
16	Following the assignment from the WMS system and selection of the level, the elevator motors are activated.	F_LIFT_01A and F_LIFT_02A Running.
17	When the level sensor reaches its target (ELVA.LT_A-)and the elevator base sensor (ELVA.BC-B1-BP) is activated, the elevator motors stop.	F_LIFT_01A and F_LIFT_02A Stopped.
	When the lifting conveyor is in the unloading position and the ground-level exit control is active (with no light curtain alarms present), the table transfer motor runs to lower the table.	F_PCC_04A_T Running.
	When the transfer table is down, activating the position sensor 4AT-PX-DWN, the motor stops.	F_PCC_04A_T Stopped.
	When the transfer table is down and the motor is stopped, the lift carrier F_PCC_LIFT_A and conveyors begin to move, unloading the pallet from the elevator.	F_PCC_LIFT_A , and F_PCC_05A Running
18	When the conveyor sensor 5A-SEN.002 is activated, this indicates to the WMS system that the pallet is unloaded, completing the vertical transport task of the elevator and ready for the next task.	F_PCC_LIFT_A and F_PCC_05A Stopped.

11.2. Automatic Outfeed Ground Level Conveyor Sequence Elevator A

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	To change the operation of the elevator and ground level conveyor to redundant mode or flow change, the operator must enter the ELEVATORS screen on the HMI and select the flow direction from the Infeed to Outfeed button.	Conveyor and WMS in redundant outfeed mode on request

Step	Action	Completion criteria
2	If the elevator and ground level conveyor are in operation or running a task or sequence and they are transporting pallets, the system or elevator must first complete the previous task before changing the flow direction of the pallets.	Conveyor and WMS in redundant outfeed mode on request
3	When the PLC and the WMS evaluate that there are no more pallets being transported to the storage area, the flow change is carried out in the elevator and the ground level conveyor.	Conveyor in redundant outfeed mode running.
4	Before starting, the operator must verify that the conveyor has enough space to receive the pallet. To do this, it is suggested to free up space by removing the pallet from the conveyor when it is stopped to prevent operational risks.	F_PCC_01A and F_PCC_02A Stopped.
5	When the operator is unloading a pallet, the light curtain alarm is activated to stop the conveyors, ensuring that the system will automatically stop.	LIGTHC_B-SEN.001 Activated (On)
6	As mentioned in the elevator sequence, when the elevator is in the unload position, the operator must press the button (LIGTHC_B-SEN.001) to reset the alarm, clear the light curtain status and initiate the start sequence.	Outfeed control is running
7	When the control is started, the first thing the system does is determine the position of the transfer table and lower it by running the motor.	F_PCC_04A_T Running
8	When the transfer table is down, activating the position sensor 4AT-PX-DWN, the motor stops.	F_PCC_04A_T Stopped.
9	When the transfer table is down and the motor is stopped, the lift carrier F_PCC_LIFT_A and conveyors begin to move, unloading the pallet from the elevator.	F_PCC_LIFT_A , and F_PCC_05A Running
10	When the conveyor sensor 5A-SEN.002 and 4AT-SEN.002 is activated, this indicates to the WMS system that the pallet is unloaded, completing the vertical transport and the transfer table runs the F_PCC_04A_T motor to raise the table.	F_PCC_LIFT_A and F_PCC_05A Stopped. F_PCC_04A_T Running

Step	Action	Completion criteria
11	When the table rises, confirmed by the sensor 4AT-PX-UP, the motors of the transfer conveyor and the lift carrier stop, ending the process in the elevator and continuing the transport on the ground level turning on the next conveyor.	F_PCC_04A_T, F_PCC_05A and F_PCC_LIFT_A Stopped F_PCC_04A and F_PCC_03A Running
12	When the 3A-SEN.002 sensor of this conveyor is activated by the pallet, this conveyor and the previous one stop and the next zone or conveyor turns on.	F_PCC_04A and F_PCC_03A Stopped. F_PCC_02A and F_PCC_01A Running
13	The system opens ROLLER DOOR A (RD1.UP.SOL.001) when the pallet reaches its current position. When RD1-PX-UP (roller door open feedback) is triggered, the pallet continues moving.	F_PCC_03A, F_PCC_02A and F_PCC_01A Running.
14	Once the pallet has completely passed through the roller door, the system closes the door rollers RD1.DWN.SOL.001. Confirmation comes from the door closed feedback signal.	RD1-PX-DWN Activated (On)
15	When the 2A-SEN.002sensor of this conveyor is activated, the previous zone stops.	F_PCC_03A Stopped.
16	At this point, the scanner reads the pallet's barcode, informing the WMS system of the pallet that came out.	send the barcode to the WMS
17	Finally, when the last conveyor sensor is activated 1A-SEN.002, it stops the conveyors.	F_PCC_02A and F_PCC_01A Stopped.
18	Upon removing the pallet, the operator will trigger the light curtain, stopping the conveyor completely. Once the pallet is removed, the operator must reset the alarm from the local button, ending the cycle so that new pallets arriving at this point can be transported.	LIGTHC_A-SEN.001 Alarm deactivated (Off)

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11.3. Automatic Infeed Ground Level Conveyor Sequence Elevator B

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	To change the operation of the elevator and ground level conveyor to redundant mode or flow change, the operator must enter the ELEVATORS screen on the HMI and select the flow direction from the Outfeed to Infeed button.	Conveyor and WMS in redundant Infeed mode on request
2	If the elevator and ground level conveyor are in operation or running a sequence or task and they are transporting pallets, the system or elevator must first complete the previous task before changing the flow direction of the pallets.	Conveyor and WMS in redundant Infeed mode on request
3	When the PLC evaluates that there are no more pallets being transported from the storage area, the flow change is carried out in the elevator and the ground level conveyor.	Conveyor in redundant Infeed mode running.
4	As a best practice, operations personnel should first remove all pallets sent from the storage area to the ground-level conveyor. If not removed, these pallets will be returned to storage. To prevent this, it is recommended to clear all pallets from the conveyor and verify in the WMS system that no outfeed tasks remain assigned before loading new pallets into the storage section.	F_PCC_05B and F_PCC_06B Stopped.
5	Ensure the conveyor is completely stopped before placing the pallet to prevent any operational risk.	F_PCC_05B and F_PCC_06B Stopped.
6	When the operator is loading a pallet, the light curtain is triggered to stop the conveyors, ensuring the system halts in case it begins operating automatically.	LIGTHC_B-SEN.001 Alarm Activated (On)
7	The operator manually loads the pallet onto the induction conveyor.	F_PCC_05B and F_PCC_06B Stopped.

Step	Action	Completion criteria
8	From the HMI, under the INFEED/OUTFEED screen, initiate the conveyor by selecting "Start" in the Control Elve B Infeed box. The operator must also press the button to reset (START_RST-PB.002) the alarm, clear the light curtain status, and initiate the start sequence.	F_PCC_05B and F_PCC_06B Running.
9	As the conveyor starts, the sensor DIM2-SEN.005 on Dimensioner B is triggered, beginning pallet size evaluation and also the scanning of the barcode.	Dimensioner B is Testing and running.
10	As the conveyor starts, the sensor DIM2-SEN.005 on Dimensioner B is triggered, beginning pallet size evaluation and also the scanning of the barcode.	Dimensioner B is Testing and running.
11	The evaluation ends when the DIM2-SEN.005 no longer detects the pallet, when any dimensioner sensor detects an oversize condition, or when the barcode reading error occurs.	Dimensioner B Stopped.
12	Once dimensioning completes, the barcode scanner is triggered and sends the pallet data to the WMS, which returns a location assignment.	WMS Response
13	If the result is negative (either from dimensioning or WMS response): the Infeed/Outfeed control stops. The operator is then given the option to return the pallet by pressing the Start/reset button (START_RST-PB.002), which changes the direction of the conveyor and allows the pallet to be removed.	Dimensioner B, F_PCC_05B and F_PCC_06B Running.
14	Once the pallet is positioned for unloading—either for dimension adjustments or replacement with a pallet bearing a valid barcode—it is necessary to acknowledge the alarm on the Infeed/Outfeed HMI screen. After that, the Start/Reset button (START_RST-PB.002) must be pressed again to allow the pallet to be re-evaluated by the dimensioner.	Dimensioner B, F_PCC_05B and F_PCC_06B Stopped.
15	If the result is positive, the pallet continues, and the next conveyor starts running.	F_PCC_06B, F_PCC_05B and F_PCC_04B running.
16	When sensor 4B-SEN.002 is activated, stopping conveyors F_PCC_06B, F_PCC_05B and F_PCC_04B, and starting F_PCC_03B conveyor. This is reported to the WMS to assign the task to the elevator.	F_PCC_06B, F_PCC_05B and F_PCC_04B Stopped. F_PCC_03B running.

Step	Action	Completion criteria
17	The system opens theROLLER DOOR B automatically when it is detected by the storage area's entry sensor. When RD2-PX-UP (door open feedback) is triggered, the pallet continues moving.	F_PCC_04B and F_PCC_03B running.
18	After the pallet completely passes through the roller door the system automatically closes roller door RD2.DWN.SOL.002. The closed-door feedback signal confirms the operation.	RD2-PX-DWN Activated (On)
19	When the pallet reaches 3B-SEN.002, the transfer conveyor table raises.	F_PCC_03B Stopped. F_PCC_04B_T running.
20	Upon feedback signal 4BT-PX-UP (Table Up), the table lift motor (F_PCC_04B_T) stops and the pallet conveyor resumes.	F_PCC_04B_T Stopped. F_PCC_03B and F_PCC_01B running.
21	When the pallet activates sensor 4BT-SEN.002, the table lowers via its motor	F_PCC_04B_T running.
22	When the table down feedback (4BT-PX-DWN). is received: The transfer table motor and previous conveyors stop, and the last infeed conveyor (F_PCC_02B) is activated.	F_PCC_03B, F_PCC_01B and F_PCC_04B_T Stopped. F_PCC_02B running.
23	When 2B-SEN.002 detects pallet presence, the conveyor stops and informs the WMS and the Elevator A that a pallet is ready.	F_PCC_02B Stopped.
24	If the WMS confirms pallet acceptance and assigns the task to Elevator B—provided it is at Level 1, in a ready state, and operating normally—the conveyors will restart, and the pallet will be loaded into the elevator.	F_PCC_02B and F_PCC_LIFT_B running.
25	The LCB-SEN.002 sensor detects the pallet on the lift conveyor. This conveyor stops, informing the elevator it is ready for vertical transport.	F_PCC_02B and F_PCC_LIFT_B Stopped.
26	The system returns to the ready state for the next pallet cycle.	

11.4. Automatic Infeed-Elevator B Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	If the elevator is running or executing a sequence and is transporting pallets, skip to step 7. If it is not executing sequences, continue with step 2.	
2	Before starting the elevator movement, the operator must check first if the chain indicators are in OK status.	ELVB.CB-B1-CC1, ELVB.CB-B2-CC2, ELVB.CB-B3-CC3 and ELVB.CB-B4-CC4 Activated (On)
3	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVB.CB-B5-L1, ELVB.CB-B6-L2, ELVB.CB-B7-L3 and ELVB.CB-B8-L4 Activated (On)
4	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVB.TC-S1-TLS1 and ELVB.TC-S2-TLS2) and bottom (ELVB.BC-S1-BLS1 and ELVB.BC-S2-BLS2) of the elevator.	ELVB.BC-S1-BLS1 and ELVB.TC-S2-TLS2 Activated (On). ELVB.BC-S2-BLS2 and ELVB.TC-S1-TLS1 Deactivated (Off).
5	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 02 STATUS" section, press the button to enable the elevator.	Enable status running.
6	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01B-PX-RET, ESC02A-PX-SHFT and ESC02B-PX-SHFT Activated (On).

Step	Action	Completion criteria
7	To change the elevator's operation to redundant mode or flow change, the operator must enter the ELEVATORS screen on the HMI and select the flow direction from the Infeed to Outfeed button.	Elevator and WMS in redundant outfeed mode on request
8	If the lift is in operation or running a sequence and is transporting pallets, the WMS system and the lift must first complete the existing task before changing the direction of pallet flow.	Elevator and WMS in redundant outfeed mode on request
9	When the PLC and WMS evaluate that there are no more pallets or tasks to transport to the storage area, the flow is changed in the elevator.	Elevator in redundant outfeed mode running
10	The elevator can perform several types of movements, each assigned by the WMS system based on priority. These movement purposes include Pallet Pickup, Pallet Dropoff, Shuttle Pickup, and Shuttle Dropoff. In this example, the ground-level transporter scanned the pallet’s barcode and sent the information to the WMS system, which then assigned the elevator the task of Pallet Pickup.	The WMS system assigns the Pallet Pickup task
11	If the elevator is on another level, the WMS system will send the order to the elevator to go to level 1. Once the level is assigned, the elevator motors are activated.	F_LIFT_01B and F_LIFT_02B Running.
12	When the level sensor ELVB.LT_B- reaches the target and the level sensor 1 ELVB.BC-B1-BP is activated, the motors are turned off, stopping the elevator.	F_LIFT_01B and F_LIFT_02B Stopped.
13	If the WMS has confirmed pallet acceptance and Elevator B is at Level 1, Ready to load and healthy, the conveyors restart, and the pallet is loaded into the elevator.	F_PCC_02B and F_PCC_LIFT_B running.
14	The LCB-SEN.002 sensor detects the pallet on the lift conveyor. This conveyor stops, informing the and the WMS system that it is ready for vertical transport.	F_PCC_02B and F_PCC_LIFT_B Stopped.
15	Once the pallet is inside the elevator and the WMS has provided the destination level, this information is assigned to the elevator to execute the transport operation.	The WMS system assigns the Pallet Dropoff task.

Step	Action	Completion criteria
16	With the "End Stop" motors stopped and the position sensors activated, the elevator is ready to move and turns on the elevator motors, transporting the pallet to the target level.	F_LIFT_01B and F_LIFT_02B Running.
17	When the elevator level sensor (ELVB.LT_B-) detects that it has reached the target level and the lift carrier sensor (ELVB.CB-W1-CP) is activated, the elevator motors stop and the elevator carrier motors and the level it reached are activated to unload the pallet (level 4 as an example).	F_LIFT_01B and F_LIFT_02B Stopped. F_PCC_LIFT_B and F_PCC_WSA_04B Running.
18	Once the pallet is loaded on the level conveyor (Example level 4) and is detected by the front sensor (HO-4B-SEN.002) the conveyor motor will stop and inform Stow system and WMS system that the area is occupied and is ready to be unloaded by the robot.	F_PCC_WSA_04B Stopped. Signal to Stow and WMS Activated.
19	When the robot picks up the package, the area sensor is deactivated, indicating that the area is now free to receive the next pallet.	HO-4B-SEN.001 AND HO-4B-SEN.002 Deactivated (Off).

12. Automatic Mode transporting shuttles between levels Operation

The previous section detailed the standard and redundant automatic configuration in which:

- Lift A handles both pallet infeed and outfeed.
- Lift B also handles both pallet infeed and outfeed.

In addition to standard pallet movement, the system is capable of transporting shuttles between levels. However, this function is only executed when explicitly commanded by the WMS system.

The following sections outline the automated operational sequences available in both normal and redundant operation modes. In these configurations, Lift A or Lift B can dynamically carry out shuttle transfers based on system requirements.

Each sequence is executed in real time, driven by current system status, pallet position, and data from sensors, actuators, and motors controlled via Variable Speed Drives (VSDs).

As a first step, if the system is not running, it is necessary to select automatic mode and start the system.

Mode activation sequence.

Permissive list		
No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	Action	Completion criteria
1	Confirm safety contactors are energized KM010_1 and KM010_2. The system should indicate that the safety circuits are engaged and operational.	System is healthy
2	Select "Automatic Mode" on the main screen of the HMI.	System in Automatic mode
3	Press "Start" on the main HMI screen to activate the system in Automatic Mode.	System is running

12.1. Automatic Elevator A transporting shuttles between levels Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	If the elevator is running or executing a task and is transporting pallets, skip to step 7. If it is not executing sequences, continue with step 2.	
2	Before starting the elevator movement, the operator must check first if the chain indicators are in OK status.	ELVA.CB-B1-CC1, ELVA.CB-B2-CC2, ELVA.CB-B3-CC3 and ELVA.CB-B4-CC4 Activated (On)
3	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVA.CB-B5-L1, ELVA.CB-B6-L2, ELVA.CB-B7-L3 and ELVA.CB-B8-L4 Activated (On)
4	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2) and bottom (ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1) of the elevator.	ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2 Activated (On). ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1 Deactivated (Off).
5	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 01 STATUS" section, press the button to enable the elevator.	Enable status running.
6	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01A-PX-RET, ESC01A-PX-SHFT and ESC01B-PX-SHFT Activated (On).

Step	Action	Completion criteria
7	If the lift is in operation or running a sequence and is transporting pallets, the WMS system and the lift must first complete the existing task before assigning the new task to the lift.	Elevator and WMS in mode on Shuttle Pickup request
8	When the PLC and WMS evaluate that there are no more pallets or tasks for the elevator, the task is assigned to it.	Elevator and WMS in mode on Shuttle Pickup request
9	The elevator can perform several types of movements, each assigned by the WMS system based on priority. These movement purposes include Pallet Pickup, Pallet Dropoff, Shuttle Pickup, and Shuttle Dropoff. In this example, the WMS system has the information that Shuttle needs to be picked up at level 4 and will be delivered to level 6.	The WMS system assigns the Shuttle Pickup task
10	If the elevator is on a different level, the WMS system will send a command to move it to the level where the shuttle will be loaded (e.g., level 4). Selecting the level activates the elevator's motors.	F_LIFT_01A and F_LIFT_02A Running.
11	When the elevator level sensor (ELVA.LT_A-) detects that it has reached the target level (level 4 as an example) and the elevator carrier sensor (ELVA.CB-W1-CP) is activated, the elevator motors stop	F_LIFT_01A and F_LIFT_02A Stopped.
12	Once the elevator is in position, the End-stop linear motor extends, running the motor.	F_END_STOP_LM_01A Running.
13	When the position sensor (LMC01A-PX-EXT) detects that the linear motor is extended, it stops. Once in this position, the rotary motors are activated to move the end stops.	F_END_STOP_LM_01A Stopped. F_END_STOP_01A and F_END_STOP_02A Running.
14	When the position sensors (ESC01A-PX-DWN and ESC01B-PX-DWN) detect the position of the rotary motors, they stop, and the elevator system informs the WMS system that the passage is clear to load the Shuttle into the elevator.	F_END_STOP_01A and F_END_STOP_02A Stopped. Informs the WMS that the elevator is ready to be loaded

Step	Action	Completion criteria
15	When the WMS system informs the elevator that Shuttle is loaded, the end-stop motors start running so that the elevator is ready to move.	F_END_STOP_01A and F_END_STOP_02A Running.
16	rotary motor sensors in position the linear motor turns on and retracts.	F_END_STOP_01A and F_END_STOP_02A Stopped. F_END_STOP_LM_01A Running
17	The system evaluates whether it is Ready to move using the carrier lift feedback sensors by stopping the motor. If these sensors are not activated, the "End Stop" motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	F_END_STOP_LM_01A Stopped. LMC01A-PX-RET, ESC01A-PX-SHFT and ESC01B-PX-SHFT Activated (On).
18	Once the pallet is Shuttle the elevator and the WMS has provided the destination level, this information is assigned to the elevator to execute the transport operation.	The WMS system assigns the Shuttle Dropoff task.
19	Following the assignment from the WMS system and selection of the level, the elevator motors are activated.	F_LIFT_01A and F_LIFT_02A Running.
20	When the elevator level sensor (ELVA.LT_A-) detects that it has reached the target level (level 6 as an example) and the elevator carrier sensor (ELVA.CB-W1-CP) is activated, the elevator motors stop	F_LIFT_01A and F_LIFT_02A Stopped.
21	When the elevator is at the level, the end-stop motor extends again.	F_END_STOP_LM_01A Running.
22	When the position sensor (LMC01A-PX-EXT) detects that the linear motor is extended, it stops. Once in this position, the rotary motors are activated to move the end stops.	F_END_STOP_LM_01A Stopped. F_END_STOP_01A and F_END_STOP_02A Running.

Step	Action	Completion criteria
23	When the position sensors (ESC01A-PX-DWN and ESC01B-PX-DWN) detect the position of the rotary motors, they stop, and the elevator system informs the WMS system that the passage is clear to load the Shuttle into the elevator.	F_END_STOP_01A and F_END_STOP_02A Stopped. Informs the WMS that the elevator is ready to be unloaded
24	When the WMS system informs the elevator that Shuttle is unloaded, the end-stop motors start running so that the elevator is ready to move.	F_END_STOP_01A and F_END_STOP_02A Running.
25	rotary motor sensors in position the linear motor turns on and retracts.	F_END_STOP_01A and F_END_STOP_02A Stopped. F_END_STOP_LM_01A Running
26	When the linear motor completes its travel and the position sensor detects this, the motor stops and informs the WMS system that the SS has been unloaded and the elevator is ready to perform another task.	F_END_STOP_LM_01A Stopped.

12.2. Automatic Elevator B transporting shuttles between levels Sequence

Permissive list		
The system must be running. No safety alarm should be activated. Devices that are to be run Automatic should not have any alarms activated.		
Step	**Action**	**Completion criteria**
1	If the elevator is running or executing a task and is transporting pallets, skip to step 7. If it is not executing sequences, continue with step 2.	
2	Before starting the elevator movement, the operator must check first if the chain indicators are in OK status.	ELVB.CB-B1-CC1, ELVB.CB-B2-CC2, ELVB.CB-B3-CC3 and ELVB.CB-B4-CC4 Activated (On)

Step	Action	Completion criteria
3	Second, the operator must check if the Pinlock feedback are detected or activated.	ELVB.CB-B5-L1, ELVB.CB-B6-L2, ELVB.CB-B7-L3 and ELVB.CB-B8-L4 Activated (On)
4	Third, the operator must check whether the override level indicators are in the expected or normal state, one activated and the other normally open, both at the top (ELVA.BC-S1-BLS1 and ELVA.TC-S2-TLS2) and bottom (ELVA.BC-S2-BLS2 and ELVA.TC-S1-TLS1) of the elevator.	ELVB.BC-S1-BLS1 and ELVB.TC-S2-TLS2 Activated (On). ELVB.BC-S2-BLS2 and ELVB.TC-S1-TLS1 Deactivated (Off).
5	Before starting the movement, the operator must open the HMI ELEVATORS screen and in the "ELEVATOR 02 STATUS" section, press the button to enable the elevator.	Enable status running.
6	Before the lift moves, the system evaluates whether it is Ready to move using the carrier lift feedback sensors. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	LMC01B-PX-RET, ESC02A-PX-SHFT and ESC02B-PX-SHFT Activated (On).
7	If the lift is in operation or running a sequence and is transporting pallets, the WMS system and the lift must first complete the existing task before assigning the new task to the lift.	Elevator and WMS in mode on Shuttle Pickup request
8	When the PLC and WMS evaluate that there are no more pallets or tasks for the elevator, the task is assigned to it.	Elevator and WMS in mode on Shuttle Pickup request
9	The elevator can perform several types of movements, each assigned by the WMS system based on priority. These movement purposes include Pallet Pickup, Pallet Dropoff, Shuttle Pickup, and Shuttle Dropoff. In this example, the WMS system has the information that Shuttle needs to be picked up at level 4 and will be delivered to level 6.	The WMS system assigns the Shuttle Pickup task
10	If the elevator is on a different level, the WMS system will send a command to move it to the level where the shuttle will be loaded (e.g., level 4). Selecting the level activates the elevator's motors.	F_LIFT_01B and F_LIFT_02B Running.

Step	Action	Completion criteria
11	When the elevator level sensor (ELVB.LT_B-) detects that it has reached the target level (level 4 as an example) and the elevator carrier sensor (ELVB.CB-W1-CP) is activated, the elevator motors stop	F_LIFT_01B and F_LIFT_02B Stopped.
12	Once the elevator is in position, the End-stop linear motor extends, running the motor.	F_END_STOP_LM_01B Running.
13	When the position sensor (LMC01B-PX-EXT) detects that the linear motor is extended, it stops. Once in this position, the rotary motors are activated to move the end stops.	F_END_STOP_LM_01B Stopped. F_END_STOP_01B and F_END_STOP_02B Running.
14	When the position sensors (ESC01A-PX-DWN and ESC01B-PX-DWN) detect the position of the rotary motors, they stop, and the elevator system informs the WMS system that the passage is clear to load the Shuttle into the elevator.	F_END_STOP_01B and F_END_STOP_02B Stopped. Informs the WMS that the elevator is ready to be loaded
15	When the WMS system informs the elevator that Shuttle is loaded, the end-stop motors start running so that the elevator is ready to move.	F_END_STOP_01B and F_END_STOP_02B Running.
16	rotary motor sensors in position the linear motor turns on and retracts.	F_END_STOP_01B and F_END_STOP_02B Stopped. F_END_STOP_LM_01B Running
17	The system evaluates whether it is Ready to move using the carrier lift feedback sensors by stopping the motor. If these sensors are not activated, the “End Stop” motors will move so that the carrier lift is in a safe moving position and activate the position sensors.	F_END_STOP_LM_01B Stopped. LMC01B-PX-RET, ESC02A-PX-SHFT and ESC02B-PX-SHFT Activated (On).
18	Once the pallet is Shuttle the elevator and the WMS has provided the destination level, this information is assigned to the elevator to execute the transport operation.	The WMS system assigns the Shuttle Dropoff task.

Step	Action	Completion criteria
19	Following the assignment from the WMS system and selection of the level, the elevator motors are activated.	F_LIFT_01B and F_LIFT_02B Running.
20	When the elevator level sensor (ELVA.LT_A-) detects that it has reached the target level (level 6 as an example) and the elevator carrier sensor (ELVB.CB-W1-CP) is activated, the elevator motors stop	F_LIFT_01B and F_LIFT_02B Stopped.
21	When the elevator is at the level, the end-stop motor extends again.	F_END_STOP_LM_01B Running.
22	When the position sensor (LMC01B-PX-EXT) detects that the linear motor is extended, it stops. Once in this position, the rotary motors are activated to move the end stops.	F_END_STOP_LM_01B Stopped. F_END_STOP_01B and F_END_STOP_02B Running.
23	When the position sensors (ESC02A-PX-DWN and ESC02B-PX-DWN) detect the position of the rotary motors, they stop, and the elevator system informs the WMS system that the passage is clear to load the Shuttle into the elevator.	F_END_STOP_01B and F_END_STOP_02B Stopped. Informs the WMS that the elevator is ready to be unloaded
24	When the WMS system informs the elevator that Shuttle is unloaded, the end-stop motors start running so that the elevator is ready to move.	F_END_STOP_01B and F_END_STOP_02B Running.
25	rotary motor sensors in position the linear motor turns on and retracts.	F_END_STOP_01B and F_END_STOP_02B Stopped. F_END_STOP_LM_01B Running
26	When the linear motor completes its travel and the position sensor detects this, the motor stops and informs the WMS system that the SS has been unloaded and the elevator is ready to perform another task.	F_END_STOP_LM_01B Stopped.

13. Alarm Management Philosophy

Outline how alarms are defined, prioritized, and managed:

- Criteria for generating alarms
- Alarm categorization (critical, warning, info)
- Alarm prioritization and suppression
- Operator response guidelines

13.1. Alarm Categorization

The system implements a hierarchical alarm and event notification structure to ensure proper response, operator awareness, and equipment safety. Alarms and events are categorized based on their severity and required operator actions:

1. Informational Event

- **Purpose:** Notifies the operator that the system is in a specific operational mode or executing an expected action.
- **Severity:** Low
- **Action Required:**
    - No acknowledgment required on the HMI
    - No reset required in the PLC.

2. System Event

- **Purpose:** Generated by the Siemens system to alert maintenance or engineering personnel that a non-critical, unusual condition has occurred.
- **Severity:** Low
- **Action Required:**
    - No acknowledgment required on the HMI
    - No reset required in the PLC

3. Warning Alarm

- **Purpose:** Custom-developed alarm to alert the operator of conditions that may impact performance or pose a potential operational risk.
- **Severity:** Medium

- **Action Required:**
  - No reset required
  - Acknowledgment not required (but logged in the HMI history for traceability)

4. Error Alarm

- **Purpose:** Indicates a specific device or logical condition is in error, often due to operational or equipment faults.
- **Severity:** High
- **Action Required:**
  - Must be **acknowledged** on the HMI
  - Requires a **manual reset** in the PLC

5. Critical Alarm

- **Purpose:** Indicates a severe fault or hazardous condition that threatens operational safety or equipment integrity.
- **Severity:** Critical
- **Action Required:**
  - Must be **acknowledged** on the HMI
  - Requires a **manual reset** on the HMI after the fault is resolved

13.2. Criteria for Generating Alarms and faults

Alarms are generated based on predefined conditions within the system that indicate abnormal behavior, malfunction, or potential safety risks. These conditions are continuously monitored by the PLC and HMI, and when met, trigger alarms according to their severity. The following criteria are used to generate alarms.

13.2.1. Device Faults

The alarms displayed at this stage are related to devices such as sensors, actuators, motors, or VSDs that report a fault, malfunction, or loss of signal, as detailed in the following section.

13.2.1.1. Sensors

Alarm or fault	Description	Possible causes
Signal Forcing. (Warnings)	This mode is used to simulate sensor activity	• While active, the sensor's digital input will ignore its actual state and instead reflect the forced signal. To exit this mode, open the sensor's pop-up window and manually deactivate the Signal Forcing function.
Signal Interrupted	This condition occurs when the digital input fails to receive a signal despite the sensor being activated.	• it may indicate a broken or loose sensor cable, or a faulty connection.
Misalignment sensor	In reflective sensors, they do not reflect, this condition occurs when the signal is not received or produces incorrect readings.	• This typically indicates that the photoelectric beam is not properly aligned with its reflector. To resolve the issue, physically adjust the sensor to ensure correct alignment with the reflector.
False Reflection	The sensors use reflectors to detect the presence or position of objects. However, reflective surfaces—such as pallet wrapping or certain packaging materials—can cause the infrared light beam to bounce back incorrectly, resulting in a false reading.	• Reflective packaging materials (e.g., plastic wrap) interfering with the sensor's infrared beam, leading to false detection
Encoder Error	Encoders are critical components used to determine the elevator's position and measure pallet dimensions. They help ensure accurate system operation and detect movement-related faults.	• Mechanical malfunction affecting encoder performance • Faulty or disconnected electrical wiring to the encoder • Sensor error or incorrect signal detection

13.2.1.2. Actuator.

Alarm or fault	Description	Possible causes
Actuator On fault (Error)	This alarm is triggered when no feedback is received from the actuator within a specified time after it is powered on.	• It indicates that the actuator's output signal is not reaching the device, • Possibly due to a malfunctioning sensor.
Actuator Off Fault (Error)	This alarm is triggered when expected feedback confirming the actuator is off is not received within a specified time frame.	

13.2.1.3. Motor or VSD

<table>
<tr><th>Alarm or fault</th><th>Description</th><th>Possible causes</th></tr>
<tr><td>Motor or VFD Run Fault (Error)</td><td>This fault occurs when the system issues a run command to the motor or VFD, but no feedback is received within a specified time. If feedback is not detected, the alarm is triggered to indicate a potential issue with the motor or VFD..</td><td rowspan="2">• It indicates that the motor’s output signal is not reaching the device,
• Possibly due to a VSD malfunction.
• Possibly due to interruption of communication with the VSD.</td></tr>
<tr><td>Motor or VFD Stop Fault (Error)</td><td>This fault is triggered when the system sends a stop command to the motor or VFD, but the expected stop feedback is not received within a specified time.</td></tr>
<tr><td>Motors or VSD Trip overcurrent Fault (Error)</td><td>This fault can occur when the PLC or VSD logic detects an overcurrent condition.</td><td>• Mechanical failure or a jammed/stuck shaft
• Internal fault in the VSD, triggering overcurrent protection
• Incorrect configuration or parameter settings in the VSD</td></tr>
<tr><td>Motor or VSD Trip Overtorque Fault (Error)</td><td>This fault occurs when the PLC or VSD detects an overtorque condition and activates protective logic.</td><td>• Mechanical failure or a jammed/stuck shaft
• VSD fault triggering overtorque protection
• Incorrect configuration or parameter settings in the VSD
•</td></tr>
<tr><td>Profinet Communication Error (Error)</td><td>This error occurs when the PLC is unable to communicate with the VSD, preventing motor control.</td><td>• Communication cable is disconnected or damaged
• One or more interconnected VSDs are powered off
• Fault in the VSD’s communication module
• Incorrect VSD configuration
• VSD is turned off</td></tr>
</table>

13.2.2. Safety System Violations

Within the PLC, safety logic takes precedence over all other processes. If any safety-related component—such as the Safety PLC—detects a discrepancy in safety I/O, internal diagnostics, an activated Emergency Stop (E-Stop), an open safety door or guard during operation, or a de-energized safety contactor when it should be active, the safety logic is triggered.

When this occurs, the system deactivates the main contactors, cutting power to the relevant components. Using the Profinet communication protocol with integrated Profisafe functionality, the system can issue emergency stop commands that immediately halt variable speed drives (VSDs) and motors without applying a deceleration ramp, ensuring a fast and safe shutdown.

Signals Monitored and Managed in the Safety System Include:

Alarm or fault	Description	Possible causes
Error in the PLC safety modules. (Critical)	This condition indicates that one or more safety modules are in an error or alarm state.	• This typically occurs due to improper intervention or tampering with the electrical connections of the safety modules, resulting in a fault condition.
Emergency Stop (E-Stop) (Critical)	As the name implies, the Emergency Stop is an electromechanical button that, when pressed, triggers a safety response that immediately halts the entire system.	• The E-Stop button has been manually pressed • The connection cable is damaged or improperly connected
Door Switches (Critical)	Door switches are safety devices designed to protect operators and personnel. Since storage areas contain many moving components, these switches are installed on access doors to prevent accidental exposure to hazardous motion. When a door is opened while the system is in operation, the switches trigger an immediate system shutdown.	• The system is in automatic or semi-automatic mode, and a door has been opened, activating the safety circuit • The connection cable is damaged or improperly connected • The mechanical alignment of the switch is incorrect, preventing the safety circuit from closing properly.
Light Curtains A (Warnings)	Light curtains are safety sensors designed to detect unauthorized or unexpected presence at the system's entry or exit points. They play a critical role in preventing accidents by halting system operation when the protected area is breached.	• Power beams are blocked or have been interrupted, causing the transport system to stop. *(Once the area is cleared, the operator must reset the alarm to resume operation.)* • The connection cable is damaged or improperly connected

13.2.3. Process Deviations

This section addresses common errors or faults that may occur during system operation. These deviations typically indicate that a component has not completed its task within the expected timeframe or logical sequence.

The specific alarms associated with these deviations are described in the following section.

Alarm or fault	Description	Possible causes
Shuttle - Error - Refer MOVU HMI (Error)	The system communicates with the MOVU system and can relay shuttle status or error messages to the elevator or WMS. This error message indicates an issue reported by the external system.	• This error is triggered when the MOVU system or WMS notifies the PLC that Shuttle is in a fault or error state... *Refer to the MOVU or WMS system manuals for detailed diagnostics and resolution steps.*
Elevator - Chain Error (Critical)	The elevator operates using chains to perform vertical movement. If these chains are improperly installed or insufficiently tensioned, it can lead to poor performance or severe mechanical failure.	• Incorrect chain installation or improper tensioning • Sensor failure or misdetection related to chain position or movement • The connection cable is damaged or improperly connected
Elevator - Locking Pin Error (Critical)	The elevator is equipped with locking pins to secure it in a safe position during movement and while stationary. A locking pin error indicates a malfunction in this safety mechanism.	• Mechanical failure in the locking pin system • Actuator is not receiving the electrical signal or is mechanically stuck • Sensor failure or incorrect detection of the pin's position • The connection cable is damaged or improperly connected
Elevator Not At Level (Error)	Elevator position is monitored by two types of sensors: • **Analog sensor:** Continuously reports the elevator's position using variable values. • **Digital sensor:** Confirms whether the elevator is precisely aligned with a specific level. This alarm is triggered when there is a mismatch between the readings from these two sensors.	• Mechanical malfunction affecting elevator positioning • Misalignment or incorrect installation of the target plate for the digital sensor • Electrical connection issues with either sensor • Faulty or inaccurate sensor detection • The connection cable is damaged or improperly connected

Alarm or fault	Description	Possible causes
Paddle Stop Error – Rotation (Error)	To enable shuttle transport between levels, the end stops must be disengaged. This is done by rotating a paddle mechanism, whose position is monitored by a series of sensors. If the paddle does not reach the expected position within a predefined time, this alarm is triggered.	• Mechanical malfunction preventing proper paddle rotation • Misalignment or incorrect placement of the sensor target (stage or reflector) • Electrical connection issues with the sensors • Sensor malfunction or failure to detect paddle position
Paddle Stop Error – Retraction (Error)	To enable the shuttle to move between levels, the end stops must be deactivated. This is accomplished by extending or retracting the paddle mechanism, whose position is monitored by a series of sensors. If the paddle fails to reach its intended position within a specified time frame, this alarm is triggered.	
Paddle Stop Error – Extension (Error)		
Pallet Not On Conveyor (Error)	If any of the Paddle Stop motors are kept running for a configured time, this alarm will be activated, informing that there is an error.	
Dimension Fail – Barcode (Warnings)	This alarm is triggered when the dimensioner detects a pallet or barcode that is outside of acceptable parameters or cannot be correctly processed.	• The pallet exceeds allowable size limits • A sensor is out of its operating range • A sensor is providing incorrect measurements • The barcode is illegible and could not be read • The barcode is not assigned to the correct storage location • Scanner malfunction or communication error •
Pallet not detected at expected	A pallet not being detected at the expected sensor location within the defined timeout period	•

14. References.

List any standards, technical manuals, or other documents referenced:

- ISA-5.1: Instrumentation Symbols and Identification.
- IEC 61511: Functional Safety – Safety Instrumented Systems.
- Control system design manuals.